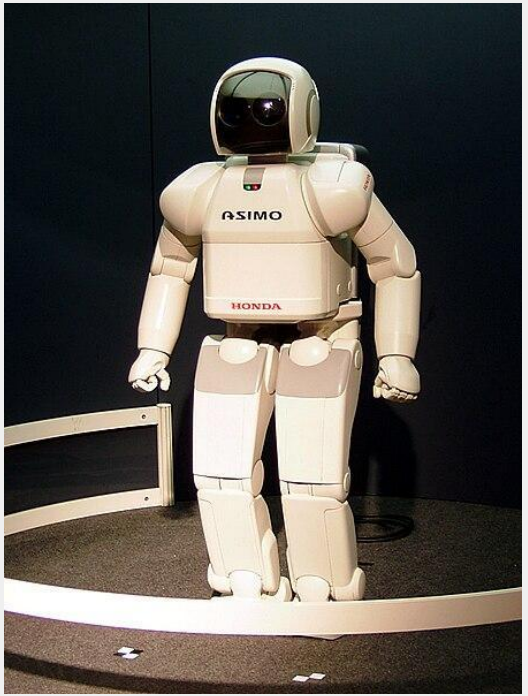
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 9 March 21, 2025

NVIDIA

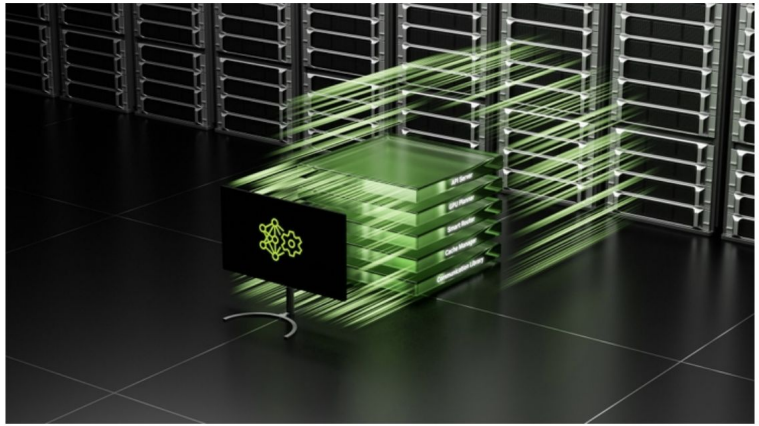
Newsroom | NVIDIA in Brief | Exec Bios | NVIDIA Blog | Podcast | Media Assets | In the N

Press Release

NVIDIA Dynamo Open-Source Library Accelerates and Scales AI Reasoning Models

NVIDIA Dynamo Increases Inference Performance While Lowering Costs for Scaling Test-Time Compute; Inference Optimizations on NVIDIA Blackwell Boosts Throughput by 30x on DeepSeek-R1

March 18, 2025



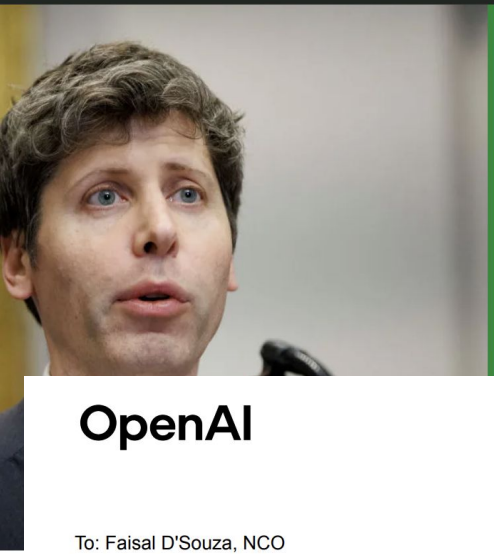
GTC—NVIDIA today unveiled NVIDIA Dynamo, an open-source inference software for accelerating and scaling AI reasoning models in AI factories at the lowest cost and with the highest efficiency.

Efficiently orchestrating and coordinating AI inference requests across a large fleet of GPUs is crucial to ensuring that AI factories run at the lowest possible cost to maximize token revenue generation.

"new AI inference-serving software designed to maximize token revenue generation for AI factories deploying reasoning AI models. It orchestrates and accelerates inference communication across thousands of GPUs, and uses disaggregated serving to separate the processing and generation phases of large language models (LLMs) on different GPUs."

"Using the same number of GPUs, Dynamo doubles the performance and revenue of AI factories serving Llama models on today's NVIDIA Hopper™ platform. When running the DeepSeek-R1 model on a large cluster of GB200 NVL72 racks, NVIDIA Dynamo's intelligent inference optimizations also boost the number of tokens generated by over 30x per GPU."

TechCrunch Latest Startups Venture Apple Security AI Apps SXSW 2025 | Events Podcasts Newsletters Sig



AI

OpenAI calls DeepSeek 'state-controlled,' calls for bans on 'PRC-produced' models

OpenAI

March 13, 2025

To: Faisal D'Souza, NCO
Office of Science and Technology Policy
2415 Eisenhower Avenue
Alexandria, VA 22314

From: Christopher Lehane
OpenAI
1455 3rd Street
San Francisco, CA 94158

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"It is the policy of the United States to sustain and enhance America's global dominance in order to promote human flourishing, economic competitiveness, and national security"
– President Donald J. Trump, Executive Order 14179, January 23, 2025

OpenAI respectfully submits the enclosed proposals to the Office of Science and Technology Policy as it weighs a new AI Action Plan that will, as Vice President Vance stated recently at the Paris AI Action Summit, maintain American leadership in AI and "make people more productive, more prosperous, and more free." As America's world-leading AI sector approaches artificial general intelligence (AGI), with a Chinese

To expand market share in Tier I countries, American commercial diplomacy policy should:

- Encourage cross-border capital flows and promote software frameworks that are optimized for domestic chip design.
- Coordinate global bans on CCP-aligned AI infrastructure, including Huawei chips.
- Continue to represent American company interests in safety and security standards bodies, and encourage global regulators to adopt pro-growth safety and security policies.
- Revise the existing export control rules to eliminate country caps on compute.
- Maintain existing export license exceptions (e.g., license exception ACM) that enable exports of technology and software for technical collaboration with allies and preservation of economically critical supply chains

To protect the US-developed IP needed to operate data centers in Tier I countries, security requirements could include:

- Prohibiting relationships with Tier III countries' foreign military and intelligence services, and the use of data centers to support military/intelligence missions for Tier III nations or human rights violators.
- Banning the use of PRC-produced equipment (e.g., Huawei Ascend chips) and models that violate user privacy and create security risks such as the risk of IP theft.
- Maintaining corporate control by entities headquartered in Tier I countries.
- Implementing—and constantly modernizing—cybersecurity, model weight security, and personnel security controls that ideally are globally synced and coordinated among Tier I governments.

Tier III countries—including the PRC and any other country subject to a US arms embargo—should continue to be subject to strict export controls of AI systems, including existing export controls on advanced chips. The strategy could also expand established controls, for example, to include advanced chips that are required for large-scale

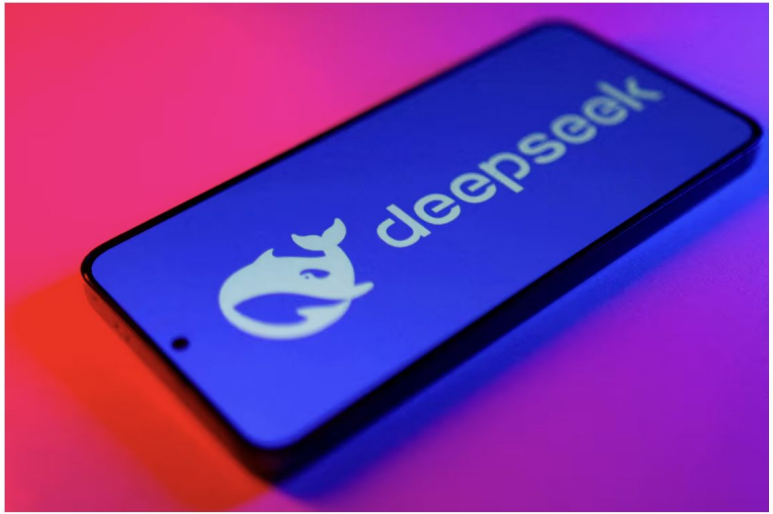
OpenAI

inference and RL training and the components used to manufacture advanced AI chips and data centers.

US Commerce department bureaus ban China's DeepSeek on government devices, sources say

By **Karen Freifeld**

March 17, 2025 7:23 PM EDT · Updated 2 days ago



The Deepseek logo is seen in this illustration taken on January 29, 2025. REUTERS/Dado Ruvic/Illustration/File Photo Purchase Licensing Rights

March 17 (Reuters) - U.S. Commerce department bureaus informed staffers in recent weeks that Chinese artificial intelligence model DeepSeek is banned on their government devices, according to a message seen by Reuters and two people familiar with the matter.

"To help keep Department of Commerce information systems safe, access to the new Chinese based AI DeepSeek is broadly prohibited on all GFE," said one mass email to staffers about their government-

"To help keep Department of Commerce information systems safe, access to the new Chinese based AI DeepSeek is broadly prohibited on all GFE,"

AI History



MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
					1	2
3	4	5	6	7 remote class + guest lecture	8	9
10	11	12	13	14 normal class ✓		16
17	18	19	20	21 normal class ✓		23
24	25	26	27	28 midterm	29	30
31				spring break		

Last Week!

- How geopolitical factors influence chip development: e.g. Taiwan during Cold War
- Outsourcing chipmaking to Asia for cheaper manual work → Asia now making most chips in the world
- Graphics cards for AI start of NPU era after AlexNet

2020s: GPUs for AI

NVIDIA Blackwell GPUs Sold Out: Demand Surges, What's Next?

Published October 14, 2024 |  Written by **Esther Shein**

The supply of next-gen processors is sold out for the next 12 months as company demand for more sophisticated AI models and applications continues unabated.

Nvidia continues to ride the AI wave as the company sees unprecedented demand for its latest Blackwell GPU next-generation processors. The supply for the next 12 months is sold out, Nvidia CEO Jensen Huang told Morgan Stanley analysts during an investors' meeting.

A similar situation occurred with Hopper GPUs several quarters ago, Morgan Stanley Analyst Joe Moore pointed out.

Nvidia's traditional customers are driving the overwhelming demand for Blackwell GPUs, including major tech giants such as AWS, Google, Meta, Microsoft, Oracle, and CoreWeave. Every Blackwell GPU that Nvidia and its manufacturing partner TSMC can produce over the next four quarters has already been purchased by these companies.

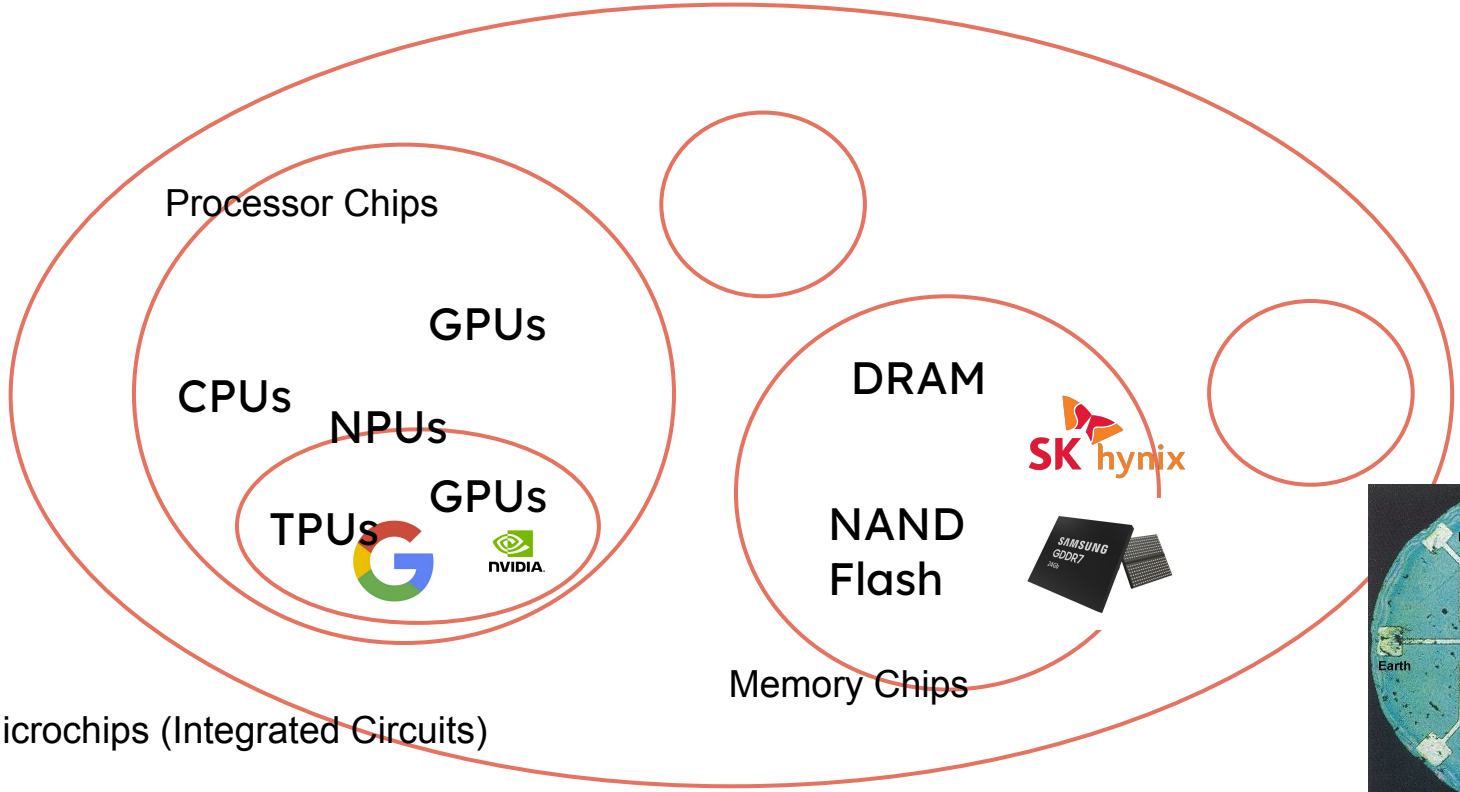
The AI Boom Is Giving Rise to "GPU-as-a-Service"

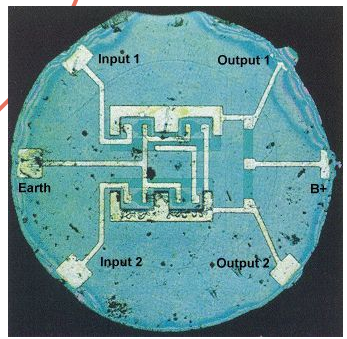
The industry harvests idle compute for AI startups that need it

BY JUAN PABLO PEREZ | 20 JAN 2025 | 3 MIN READ

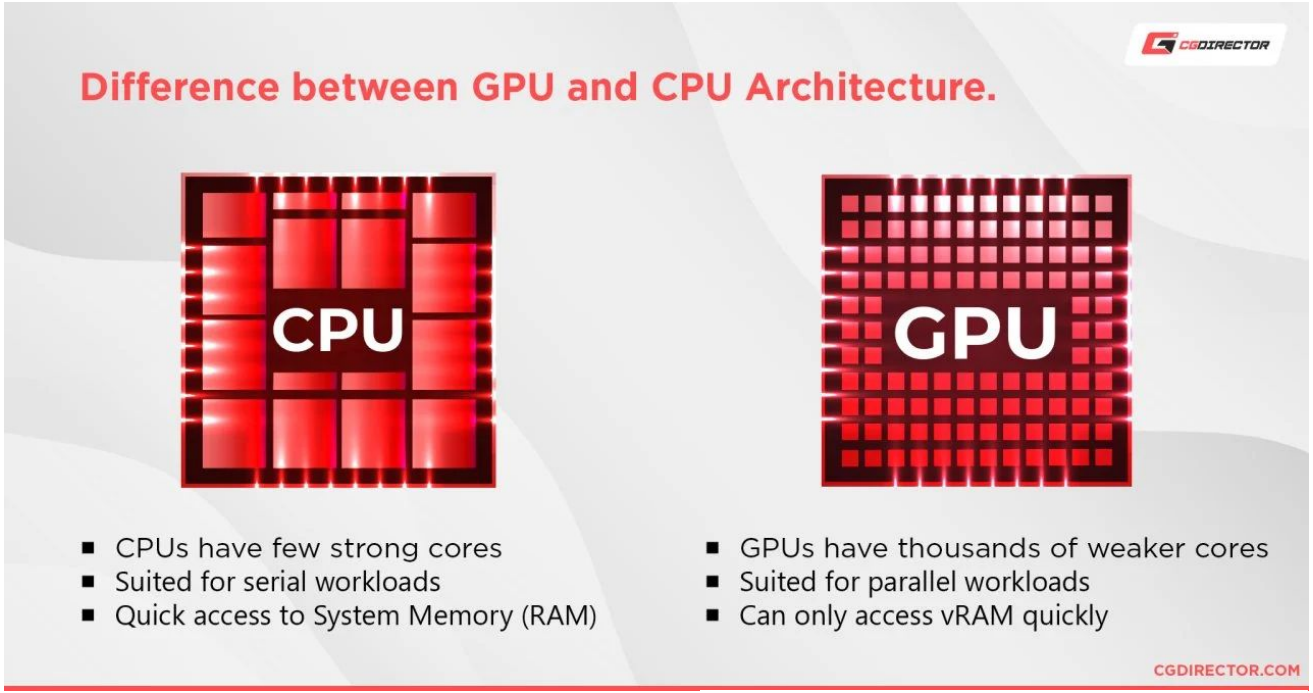


Microchips Family





Fairchild IC



GPUs

- better for batch processing
- graphics, deep learning (matrix multiplications)

CPUs

- typical PC microchips
- fast and good with general purpose computing

Chip Production

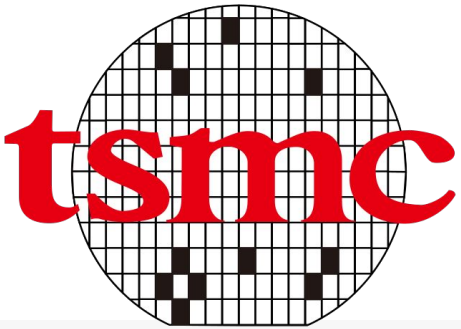
Lithography Machines

Canon

ASML

Fabrication Factories

intel



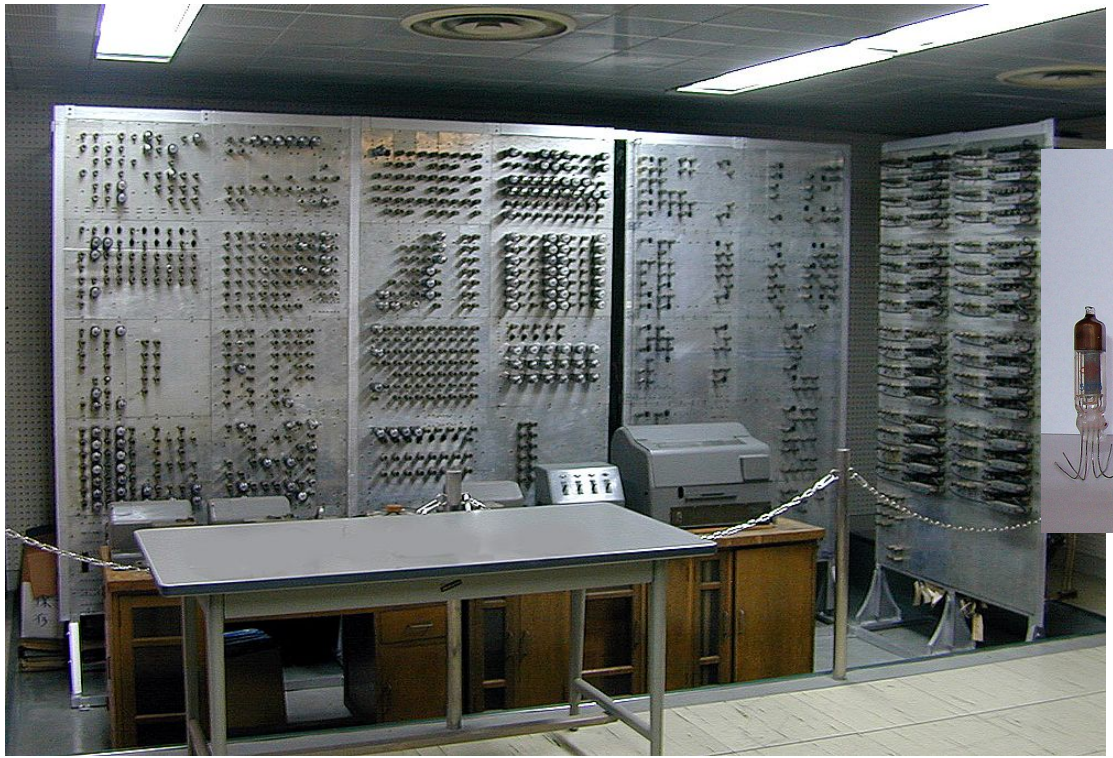
SAMSUNG

Chip Design + Software

NVIDIA

Qualcomm

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)





ENIAC 1945

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

Researchers at Bell Labs and AT&T start making semiconductors



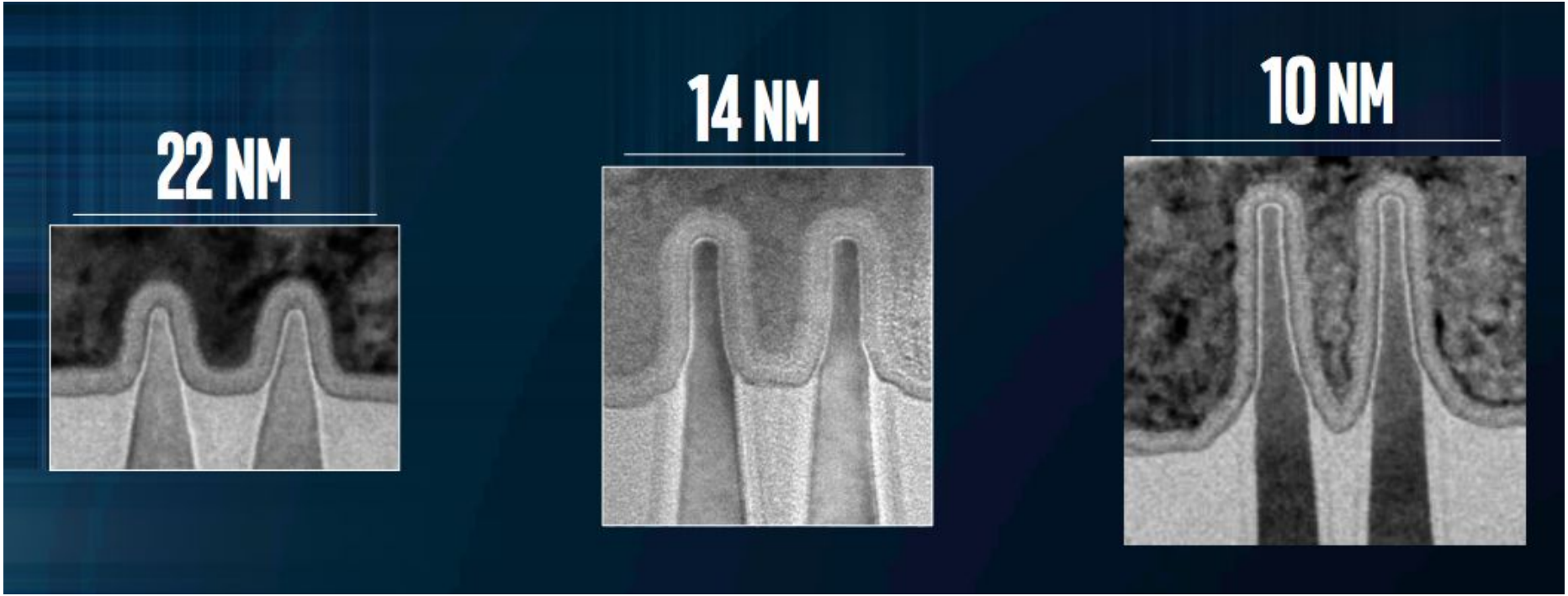
Silicon



Phosphorous

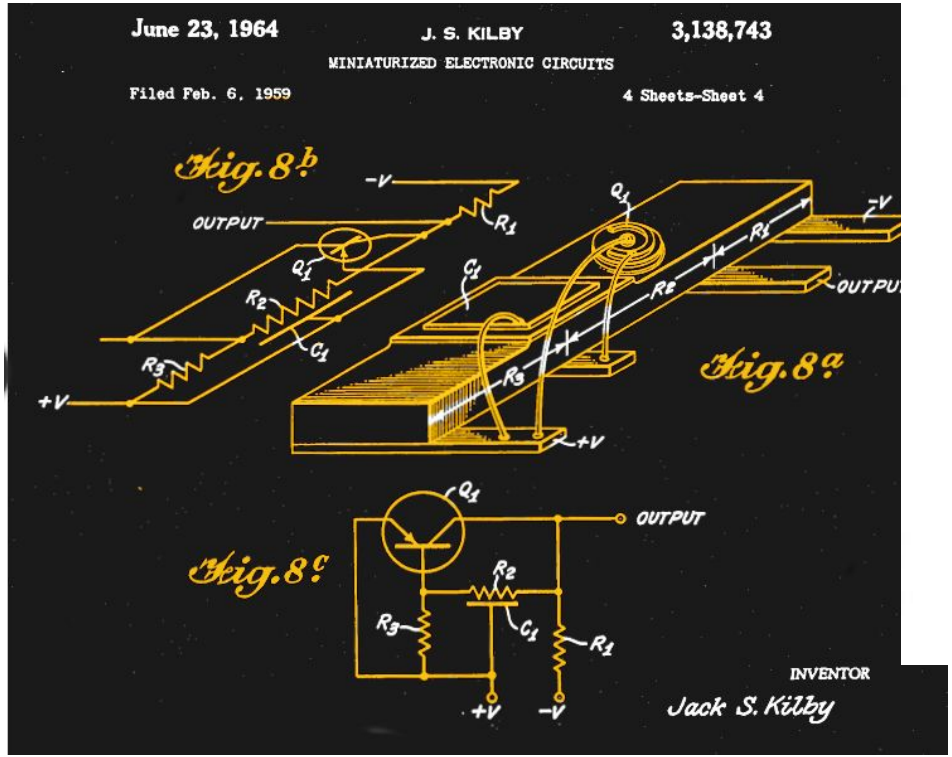
Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

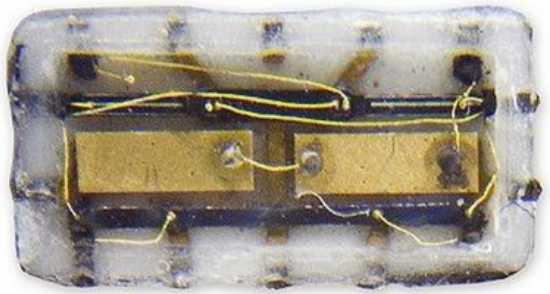
Each transistor is nanometers wide. Each chip carries billions of transistors



First Integrated Circuits 1958

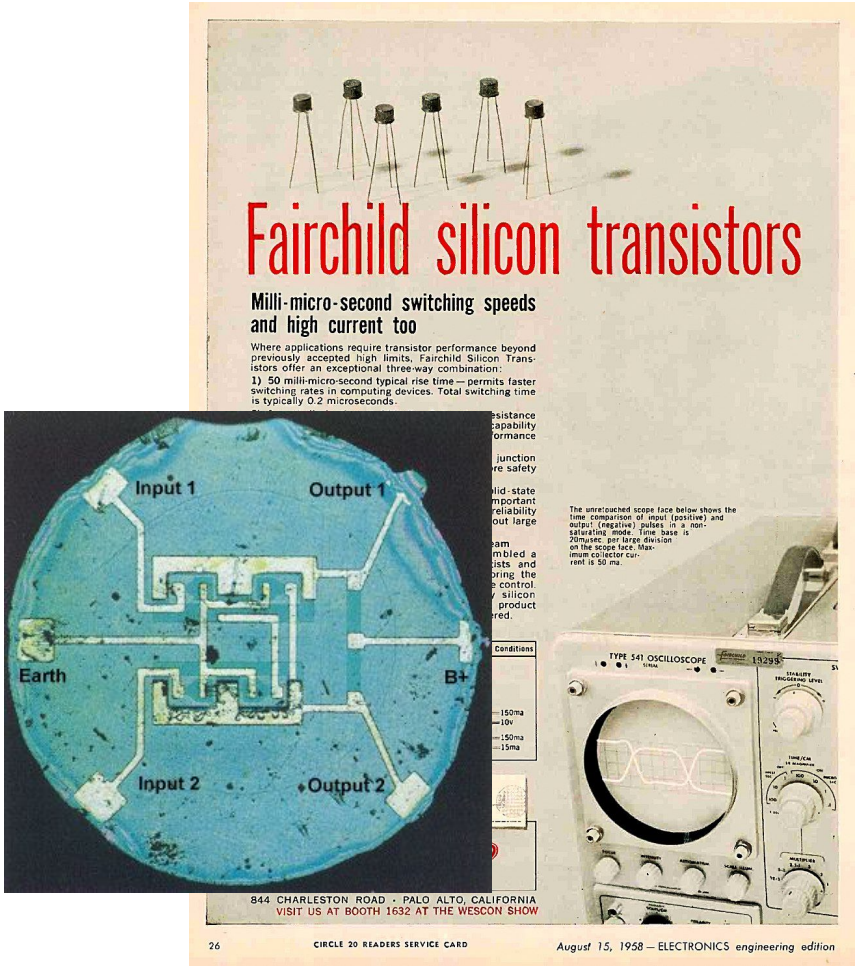
Kilby at Texas Instruments invents the Integrated Circuit





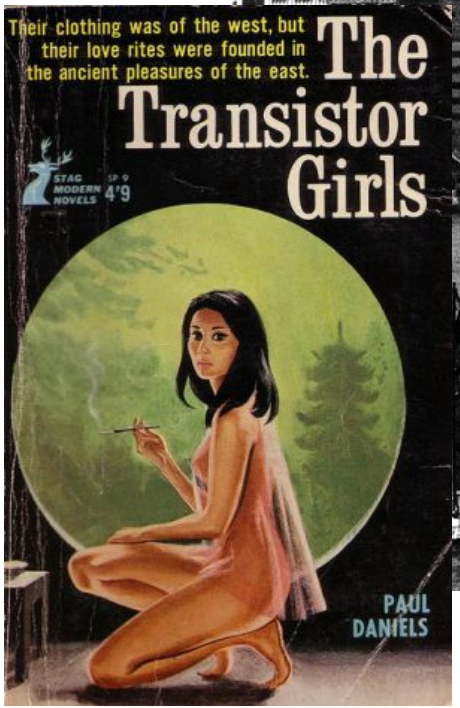
Fairchild Semiconductors

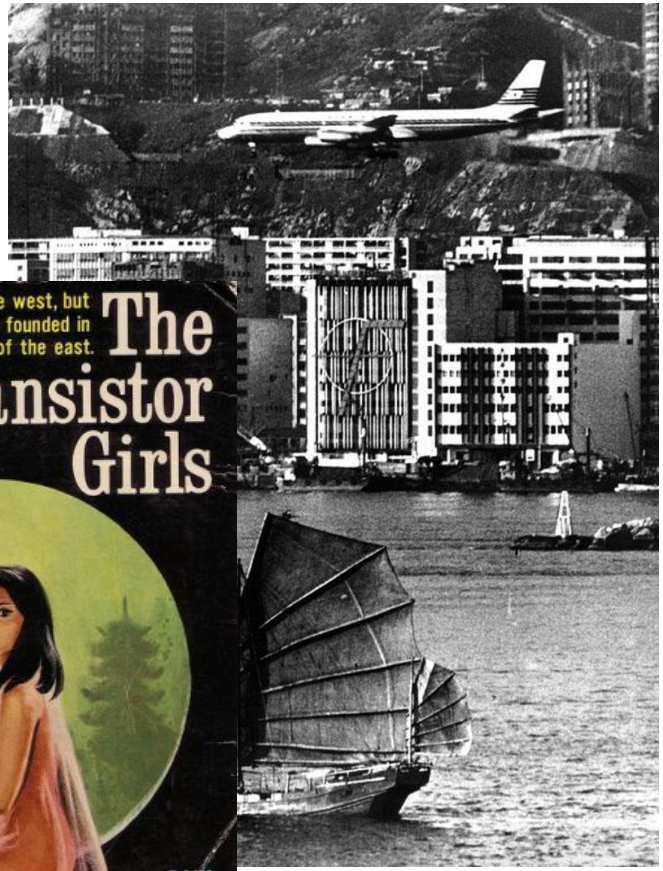
- Micrologic Chips 1958
- Apollo (NASA) program used Fairchild chips!
- Sales ballooned from $500,000 1958 to $21m two years later
- Founder Noyce started slashing prices to address wider consumer need ($120 → $15)
- 1965: military applications made up 95% of chips



Cheaper labor in Asia 1960s

- Companies like Fairchild Semiconductor, Texas Instruments, Motorola start building factories in Hong Kong, Singapore, Malaysia
- Workers here were paid about 1/10 the wages of Americans
- HK 25c (/hour), Taiwan 19c, Malaysia 15c, Singaporean 11c, South Korean 10c
- Mostly women: “better on dexterity tests”





Fairchild on Kowloon

post-Vietnam War Political Landscape

- With the loss of the U.S. in the Vietnam War looming neighboring Asian countries look to U.S.-alliance
- China's first atomic weapons test 1964
- Taiwan in particular was concerned that the fall of South Vietnam would mean Communist strengthening in the region and a larger Communist Chinese threat.

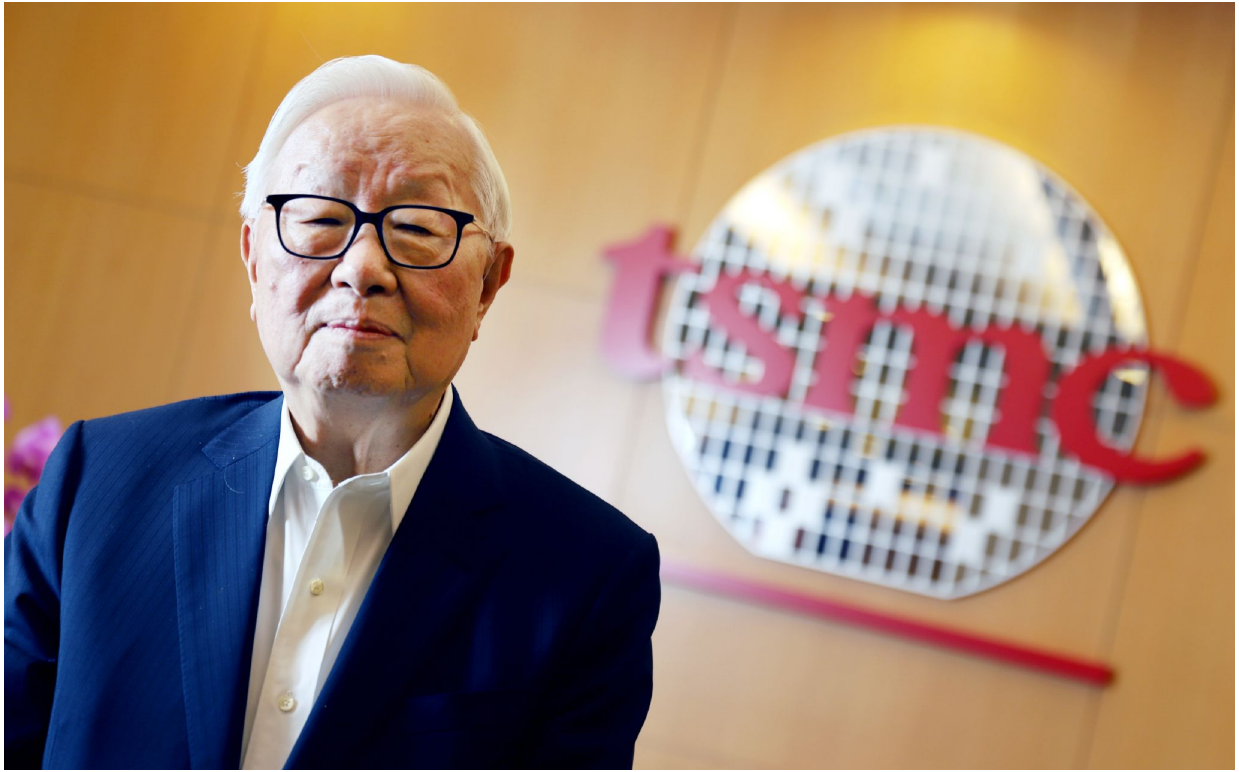
- Taiwan turned to strategic alliance with the U.S. and partnered with Texas Instruments – first plant in 1969



Minister K.T. Li

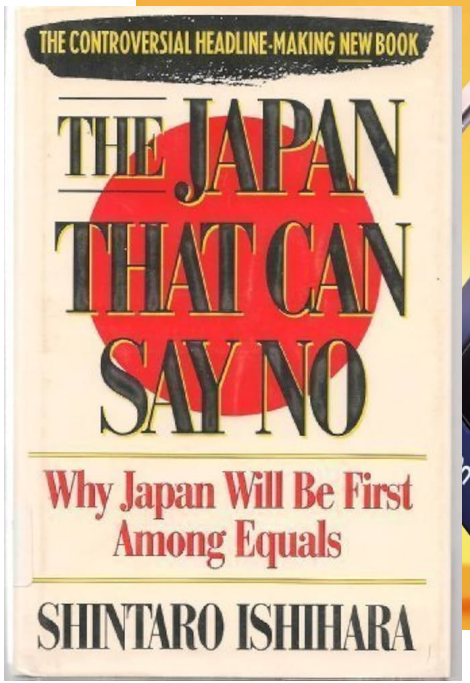
Morris Chang to lead TSMC

- Minister Li hires Morris Chang to lead TSMC in 1985
- Blank cheque from Taiwanese government (48% share)
- TSMC becomes first “foundry” – “Gutenberg for semiconductors”
- Philips gave $58m, production technology for 27.5%



Japan's productivity threatens the U.S. 1980s

- Japan's high productivity starts to threaten U.S. dominance into the 1980s
- DRAM competition with Japan weakens U.S. memory chip production
- Morita: "Militarily we could never defeat the U.S. but economically we can overcome the U.S. and become number one in the world"
- The CIA coined this the "PAX JAPONICA"



coauthored with Morita 1989



Sony Walkman 1979

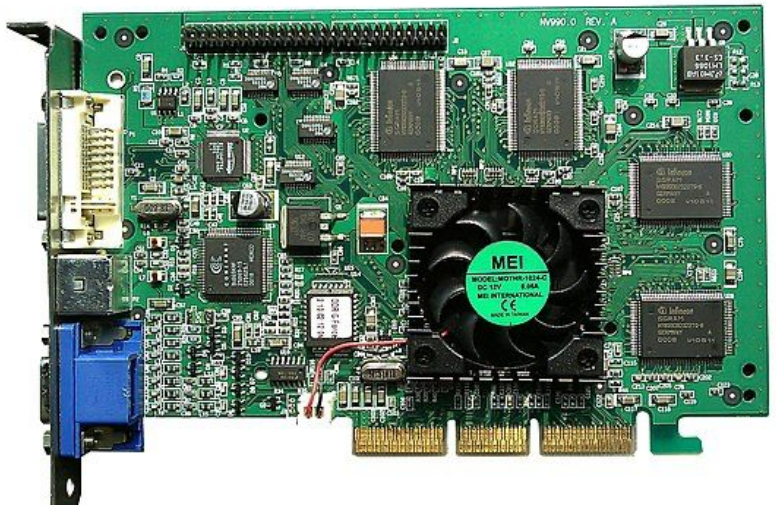
South Korean Hedge

- Silicon Valley looked to Korean producers to challenge Japanese threat
- Korean production even cheaper than Japan
- Several U.S. firms (e.g. Intel) began to sign joint ventures with Samsung in the 1980s
- Silicon Valley AMD CEO Jerry Sanders “enemy of my enemy is my friend”
- U.S. provided market and tech for Samsung chip production growth



Startups on Graphics Cards

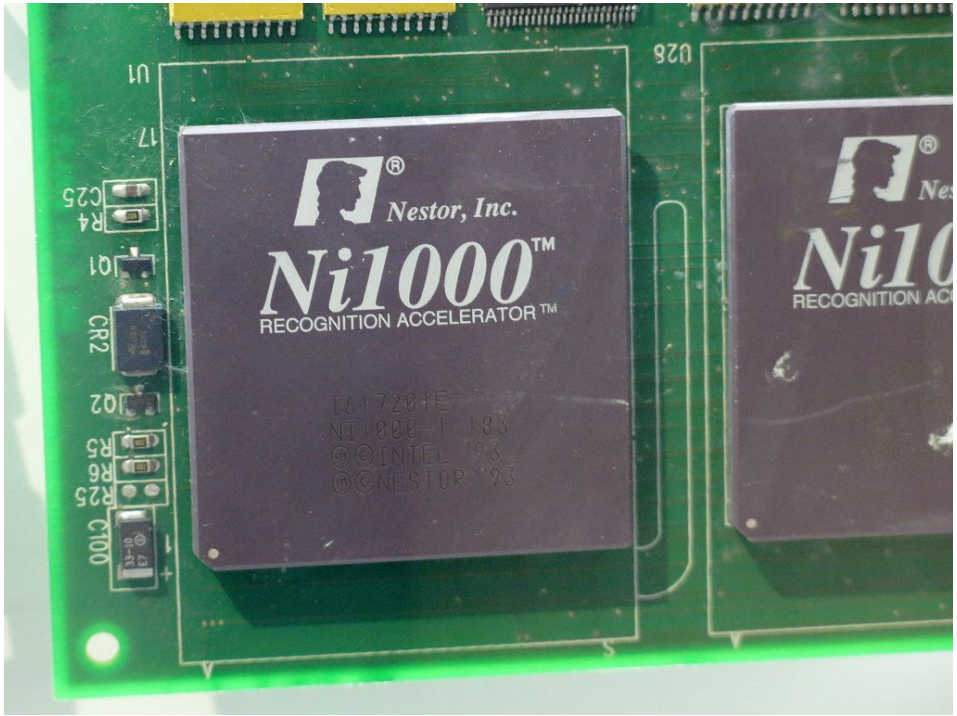
- In the 1980s Intel had a de facto monopoly on CPU microprocessors but graphics chips were a hot field for startups
- NVIDIA was founded in 1993 in Silicon Valley to design GPU chips which were mostly used for 3d graphics (for games and video)



1999 GeForce

Application Specific Integrated Circuits (ASICs)

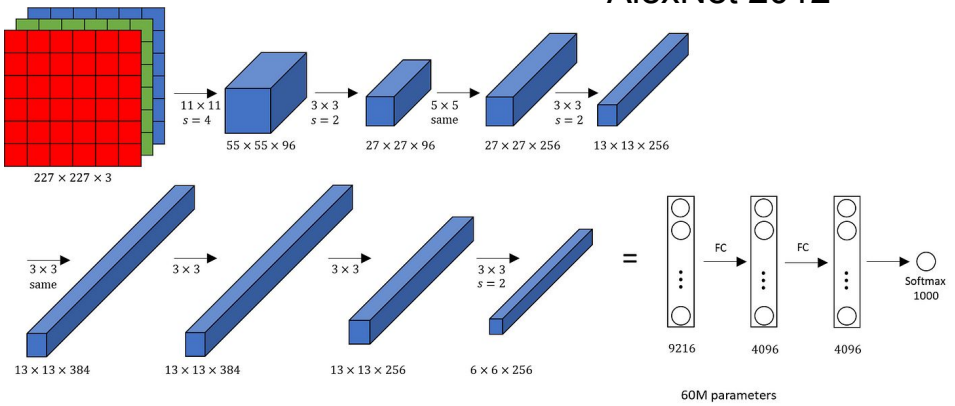
- 1980s Companies started developing microchips specifically made for neural networks
- 1980s Intel: ETANN 80170NX
- 1990s Nester/Intel: Nestor/Intel Ni1000 (DARPA) - could recognize handwritten characters
- Then 1990s, field-programmable gate arrays (FPGAs) for AI compute (away from ASICs)



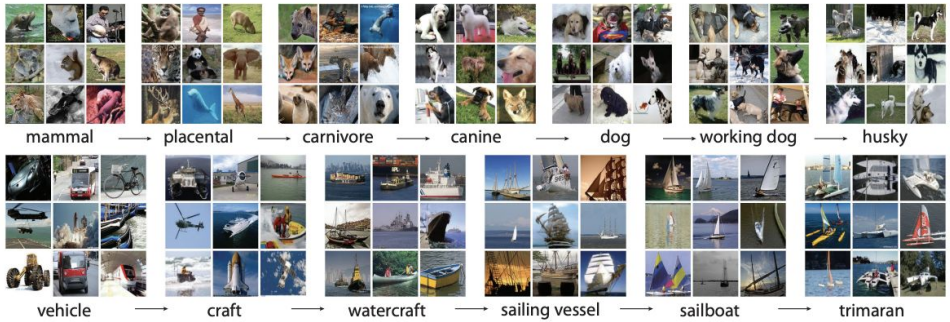
Nvidia's breakthrough? 2012 AlexNet

- AlexNet, a deep learning model proposed by researchers at U of Toronto, topped a ImageNet competition 2012 with 15% top-5 error (more than 10 percentage points lower than runner-up)
- Trained on 2 Nvidia GPUs!

AlexNet 2012



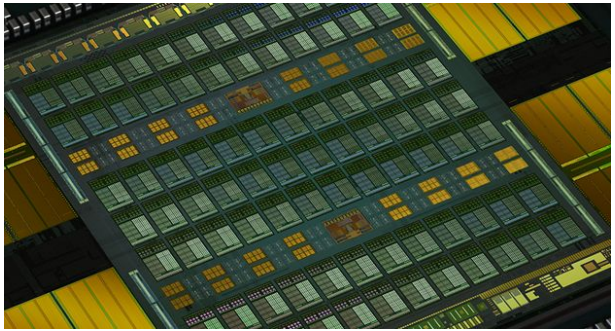
ImageNet 2009



Nvidia bets on AI future 2010 ~

- NVIDIA writes CUDA software that lets GPUS be programmed with standard programming languages (not needing graphics expertise). CUDA is only available on NVIDIA chips.
- Early 2010s Nvidia bets its future on AI hearing researchers are using GPUs not just for graphics
- NVIDIA has almost completed outsourced their manufacturing to TSMC and started focusing on software
- 2017 Tensor Cores specifically dedicated for AI/ML applications

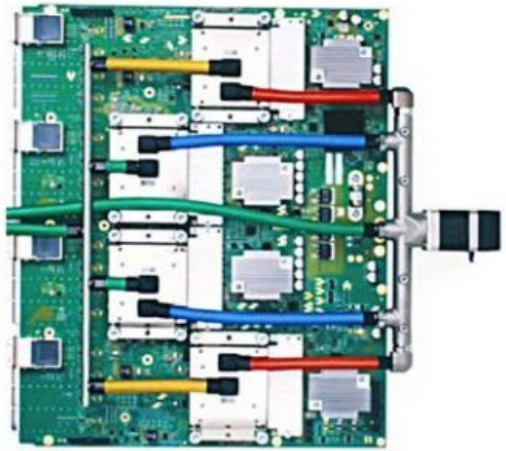




Tensor Cores in 2017 (Volta Architecture, V100 GPUs)

Big tech works on NPUs

- Other companies like Google also combined their software + hardware efforts for AI
- Google engineered the TPU (2016) to work well with Google's software Tensorflow (2015)
- Google TPU was used for AlphaGO 2016 showing magnitude higher efficiency over GPUs
- TPUs used first for Google search and Google StreetView



TPU V4



Chip Acts around the world

- August 2022: U.S. CHIPS and Science Act: $52.7 billion in subsidies and tax credits for domestic semiconductor chip production
- February 2022: European Chips Act: seeks to double the EU's share of global semiconductor production to 20% by 2030 through a €43 billion investment
- 2014, China's "Big Fund": achieve self-sufficiency in semiconductor production, third phase launched in 2024, targeting a fundraising goal of approximately $47.5 billion



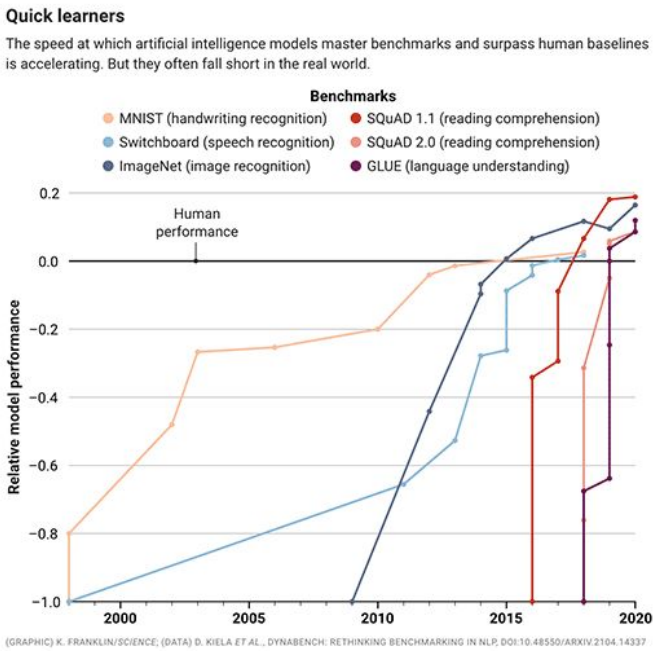


This week!

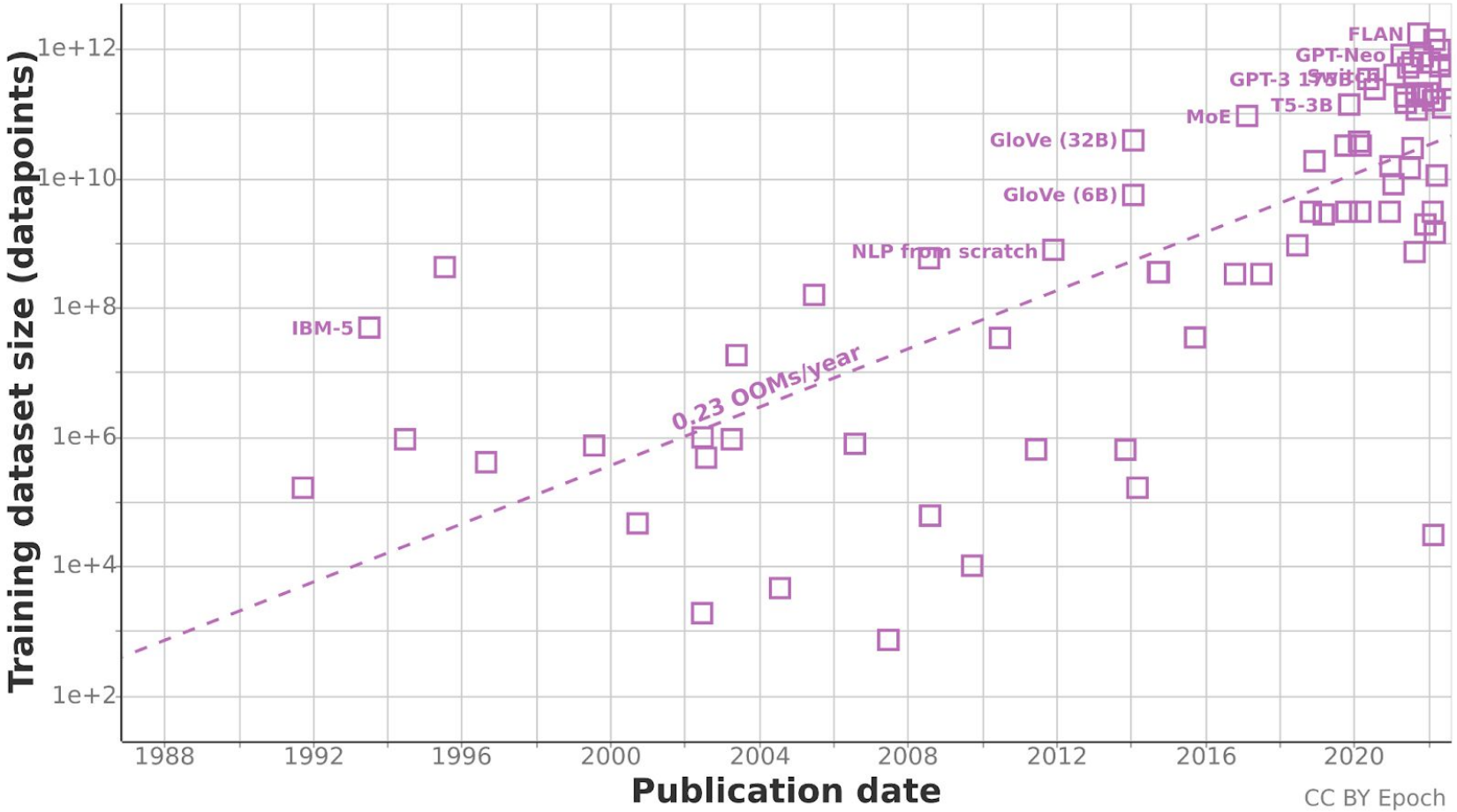
- Why data?
- What are different types of data for AI?
- How did the internet and open-sourcing culture contribute to datasets for AI?
-

Why data?

- We need data for AI because…
  - benchmarking, testing, and evaluating
  - training models



Simulated exams	GPT-4 estimated percentile	GPT-4 (no vision) estimated percentile	GPT-3.5 estimated percentile
Uniform Bar Exam (MBE+MEE+MPT)[1]	298/400 ~90th	298/400 ~90th	213/400 ~10th
LSAT	163 ~88th	161 ~83rd	149 ~40th
SAT Evidence-Based Reading & Writing	710/800 ~93rd	710/800 ~93rd	670/800 ~87th
SAT Math	700/800 ~89th	690/800 ~89th	590/800 ~70th
Graduate Record Examination (GRE) Quantitative	163/170 ~80th	157/170 ~62nd	147/170 ~25th
Graduate Record Examination (GRE) Verbal	169/170 ~99th	165/170 ~96th	154/170 ~63rd
Graduate Record Examination (GRE) Writing	4/6 ~54th	4/6 ~54th	4/6 ~54th
USABO Semifinal Exam 2020	87/150 99th–100th	87/150 99th–100th	43/150 31st–33rd
USNCO Local Section Exam 2022	36/60	38/60	24/60
Medical Knowledge Self-Assessment Program	75%	75%	53%
Codeforces Rating	392 below 5th	392 below 5th	260 below 5th
AP Art History	5 86th–100th	5 86th–100th	5 86th–100th
AP Biology	5 85th–100th	5 85th–100th	4 62nd–85th
AP Calculus BC	4 43rd–59th	4 43rd–59th	1 0th–7th
AP Chemistry	4 71st–88th	4 71st–88th	2 22nd–46th
AP English Language and Composition	2 14th–44th	2 14th–44th	2 14th–44th
AP English Literature and Composition	2 8th–22nd	2 8th–22nd	2 8th–22nd
AP Environmental Science	5 91st–100th	5 91st–100th	5 91st–100th
AP Macroeconomics	5 84th–100th	5 84th–100th	2 33rd–48th
AP Microeconomics	5 82nd–100th	4 60th–82nd	4 60th–82nd
AP Physics 2	4 66th–84th	4 66th–84th	3 30th–66th
AP Psychology	5 83rd–100th	5 83rd–100th	5 83rd–100th
AP Statistics	5 85th–100th	5 85th–100th	3 40th–63rd
AP US Government	5 88th–100th	5 88th–100th	4 77th–88th
AP US History	5 89th–100th	4 74th–89th	4 74th–89th
AP World History	4 65th–87th	4 65th–87th	4 65th–87th
AMC 10	30/150 6th–12th	36/150 10th–19th	36/150 10th–19th
AMC 12	60/150 45th–66th	48/150 19th–40th	30/150 4th–8th
Intro Sommelier (theory knowledge)	92%	92%	80%
Certified Sommelier (theory knowledge)	86%	86%	58%
Advanced Sommelier (theory knowledge)	77%	77%	46%
Leetcode (easy)	31/41	31/41	12/41
Leetcode (medium)	21/80	21/80	8/80
Leetcode (hard)	3/45	3/45	0/45


Training dataset size (datapoints)
1e+12
1e+10
1e+8
1e+6
1e+4
1e+2
1988
1992
1996
2000
2004
2008
2012
2016
2020
Publication date
FLAN
GPT-Neo
GPT-3 175B
Switch
T5-3B
MoE
GloVe (32B)
GloVe (6B)
NLP from scratch
IBM-5
0.23 OOMs/year
CC BY Epoch

Types of Data for AI Modalities

A SET OF SUPERGLUE TASKS[3]

Source: Wang et al., 2019

BoolQ	**Passage:** *Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012.* **Question:** *is barq's root beer a pepsi product* **Answer:** `No`
CB	**Text:** *B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend?* **Hypothesis:** *they are setting a trend* **Entailment:** `Unknown`
COPA	**Premise:** *My body cast a shadow over the grass.* **Question:** *What's the CAUSE for this?* **Alternative 1:** *The sun was rising.* **Alternative 2:** *The grass was cut.* **Correct Alternative:** `1`

Figure 2.3.1

HARDER QUESTIONS ADDED TO STANFORD QUESTION ANSWERING DATASET (SQUAD) 2.0

Source: Rajpurkar et al., 2018

Article: Endangered Species Act
Paragraph: "*. . . Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.*"

Question 1: "*Which laws faced significant opposition?*"
Plausible Answer: *later laws*

Question 2: "*What was the name of the 1937 treaty?*"
Plausible Answer: *Bald Eagle Protection Act*

Figure 2.3.3

Image Data



A DEMONSTRATION OF KIDNEY SEGMENTATION

Source: Kidney and Kidney Tumor Segmentation, 2021

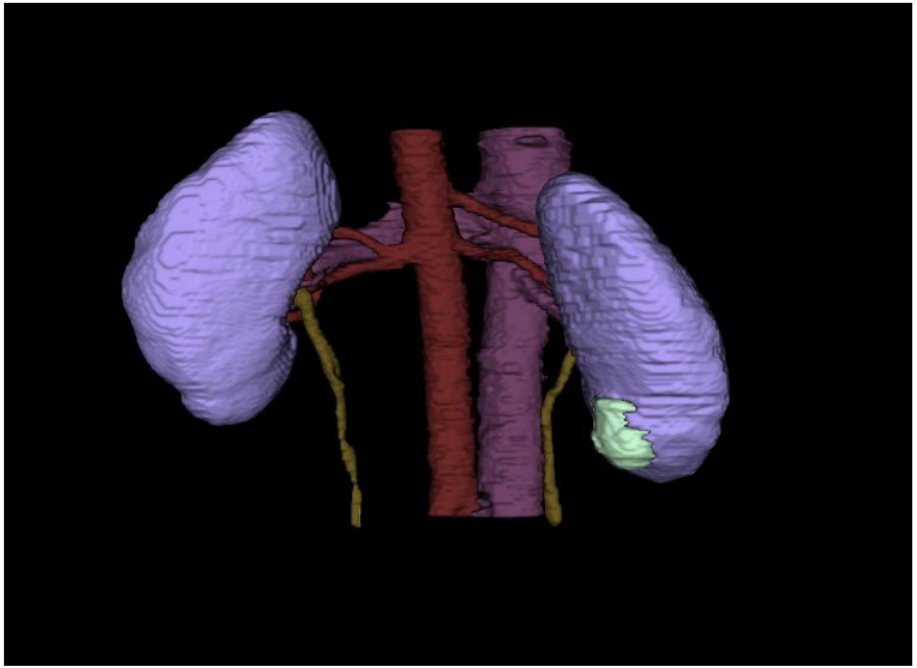


Figure 2.1.14

A DEMONSTRATION OF HOW OBJECT DETECTION APPEARS TO AI SYSTEMS

Source: COCO, 2020



Figure 2.2.4

Video Data

Video object segmentation algorithms

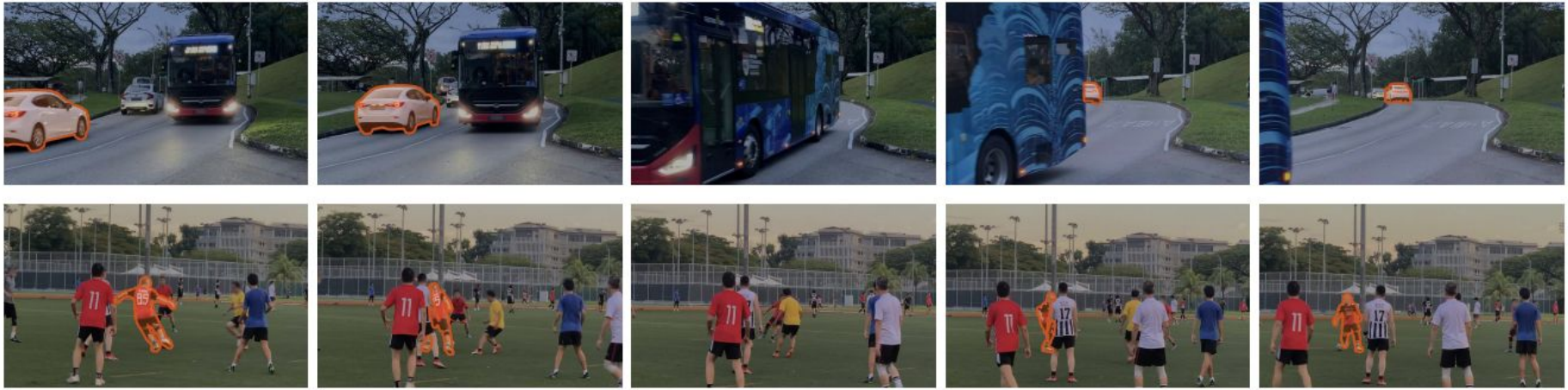
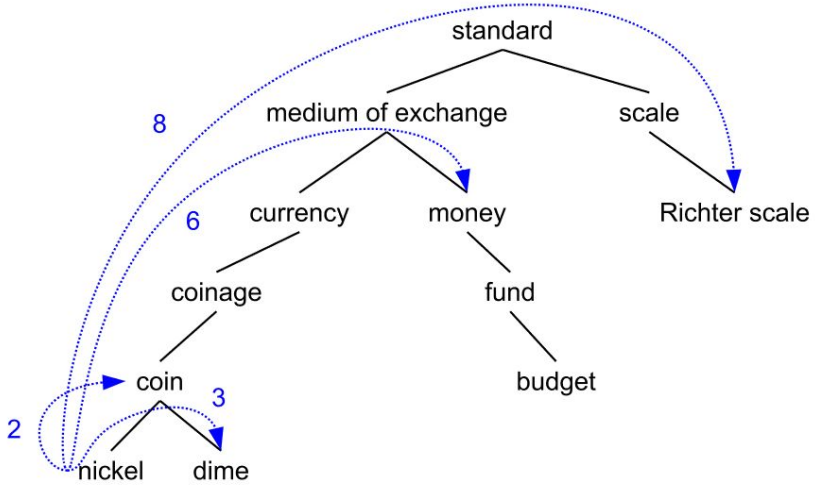
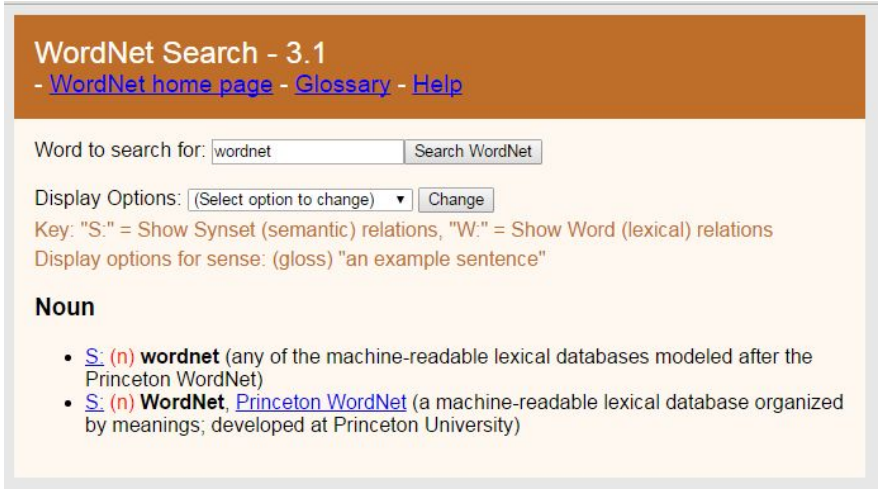


Figure 1. Examples of video clips from the co**M**plex video **O**bject **SE**gmentation (MOSE) dataset. The selected target objects are masked in orange ■. The most notable feature of MOSE is complex scenes, including the disappearance-reappearance of objects, small/inconspicuous objects, heavy occlusions, crowded environments, *etc*. For example, the target player in the 2nd row turns around when reappearing in the 4th and 5th columns after disappearing in the 3rd column, bringing challenges in re-identifying him. Most videos in MOSE contain crowded and occluded objects with the target object seldom being the salient one. The goal of MOSE dataset is to provide a platform that promotes the development of more comprehensive and robust video object segmentation algorithms.

Thanks, Internet!

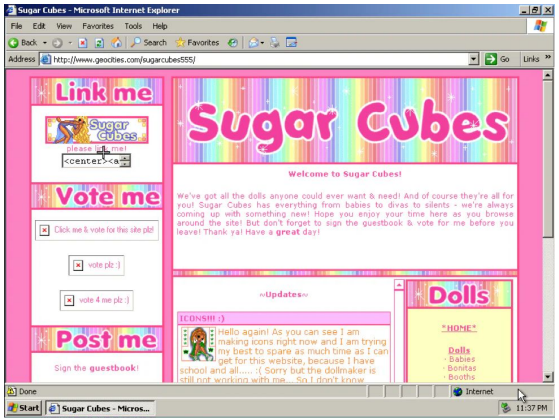
- Starting in the 90s and more faithfully into the 2000s, we get public datasets for research benchmarking and research sharing.
- Wordnet (2001) is a publicly available dictionary of 100,000 words and phrases (labeled with parts of speech tags and linked by semantic relations such a synonyms, antonyms)
- Penn Treebank (1993) is a dataset of parse trees for 3 million word corpus of English.
- British National Corpus (2001) contains 100 million English words.
- … and by the early 2000s, the World Wide Web contained several trillion words.



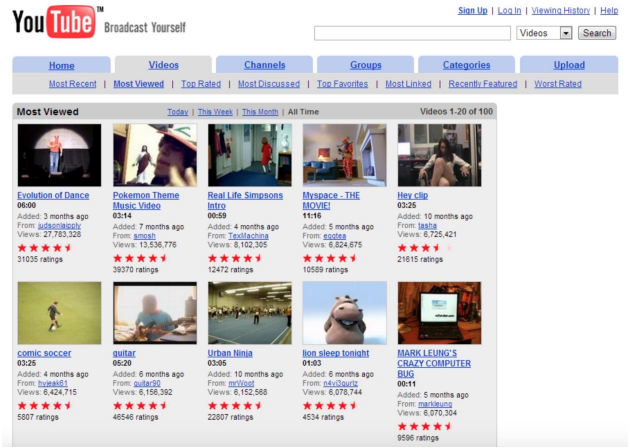


A Brief History of the Internet

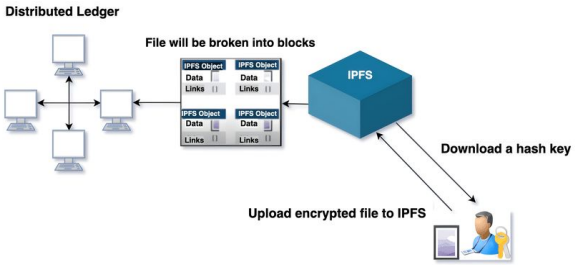
- ARPANet decommissioned 1989 but with TCP/IP protocol other parties build on it!
- Web 1.0: static web pages with little interactivity 1990s
- Web 2.0: user-generated content, dynamic platforms, social networks and real-time updates 2000s
- Web 3.0: decentralized web with crypto encryption and blockchain technology
- AI-driven, VR/AR web?



geocities



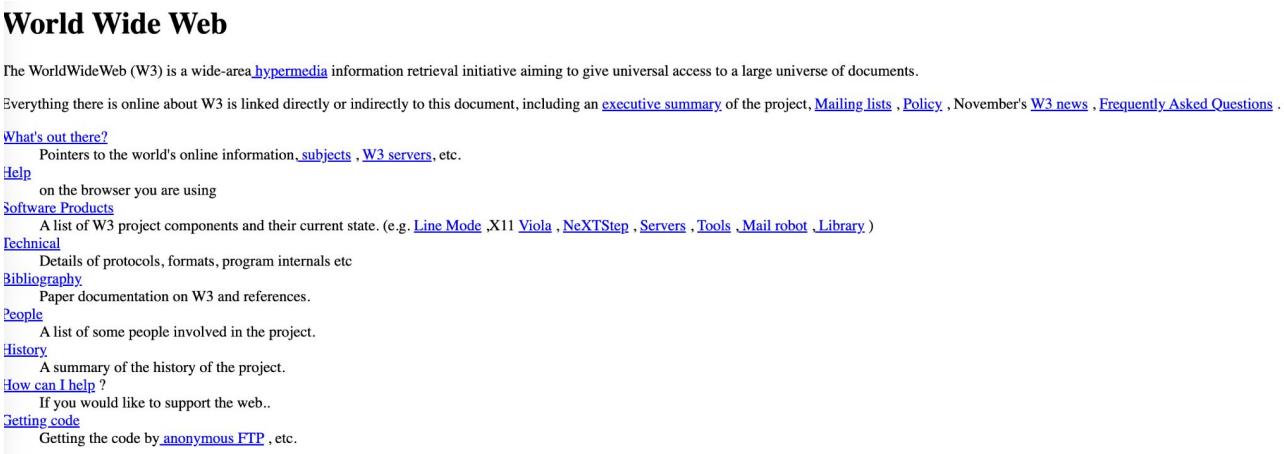
youtube



InterPlanetaryFileSystem

W.W.W., World's First Open Source Project

- ARPANet decommissioned 1989 but with TCP/IP protocol other parties built on it
- Tim Berners-Lee at CERN proposed the "web" 1989
- info.cern.ch 1991 first website ever!
- 1993 world wide web was made open source



World Wide Web

The WorldWideWeb (W3) is a wide-area hypermedia information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an executive summary of the project, Mailing lists , Policy , November's W3 news , Frequently Asked Questions .

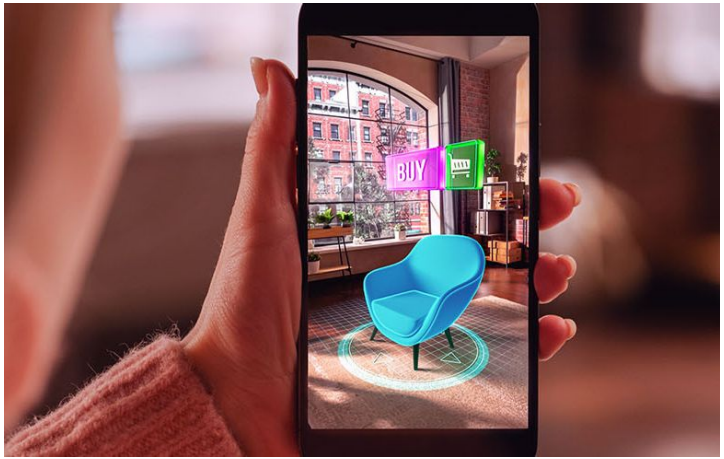
What's out there?
Pointers to the world's online information, subjects , W3 servers, etc.
Help
on the browser you are using
Software Products
A list of W3 project components and their current state. (e.g. Line Mode ,X11 Viola , NeXTStep , Servers , Tools , Mail robot , Library)
Technical
Details of protocols, formats, program internals etc
Bibliography
Paper documentation on W3 and references.
People
A list of some people involved in the project.
History
A summary of the history of the project.
How can I help ?
If you would like to support the web..
Getting code
Getting the code by anonymous FTP , etc.



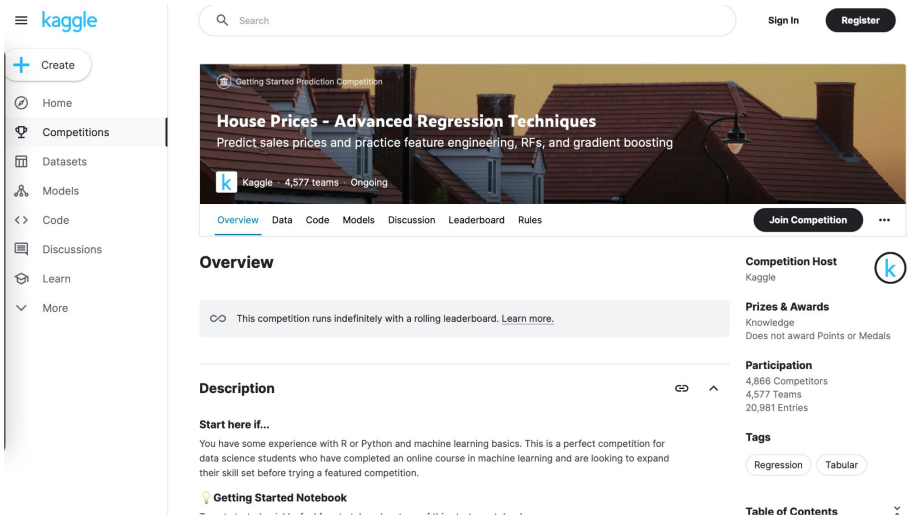
How much data online?

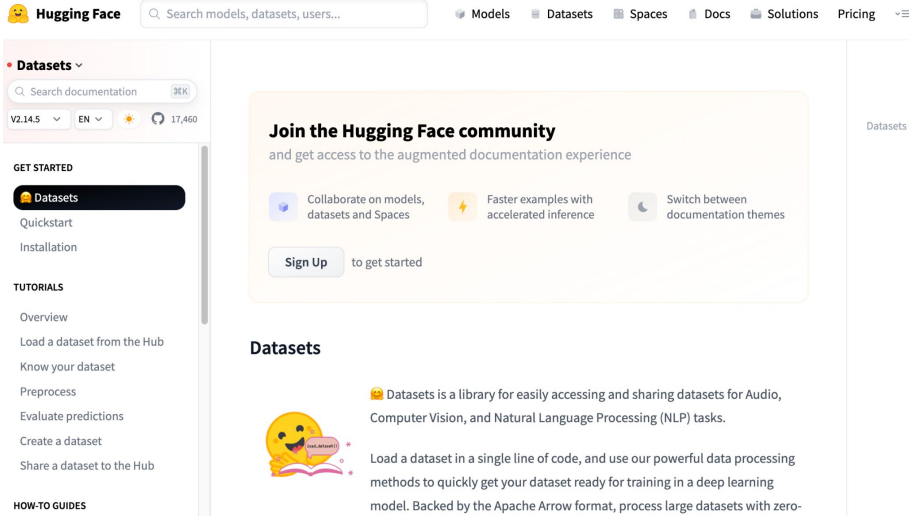
- 1990s: terabytes
- 2000s: ~10-100 exabytes
- 2010s: ~10 zettabytes
- 2020s: 100s of zettabytes?





Online data today:





Classification: Iris Dataset 1936

- Biologist & Statistician Ronald Fisher
- Dataset of Irish flower feature measures
- “The use of multiple measurements in taxonomic problems”
- 50 samples of 3 species
- Gaspe Peninsula in Canada
- 4 features: length and width of sepals and petals (cms)
- LDA model to classify (linear discriminant model)

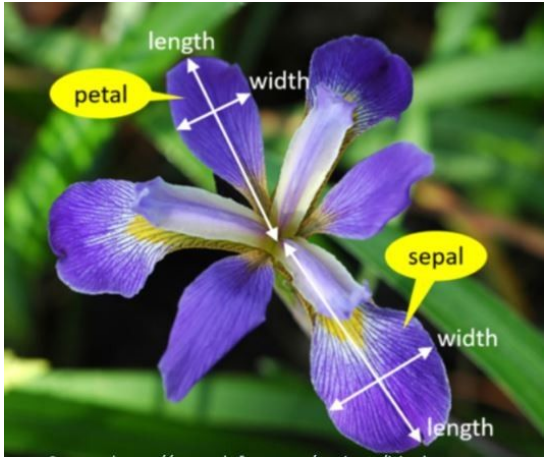


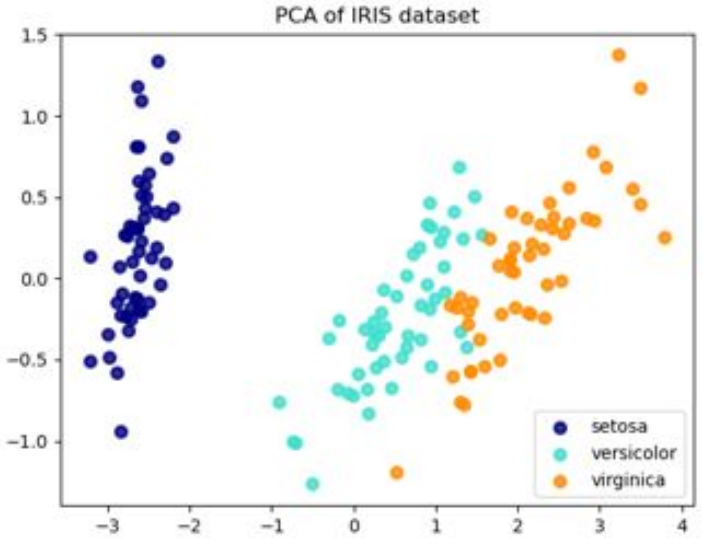






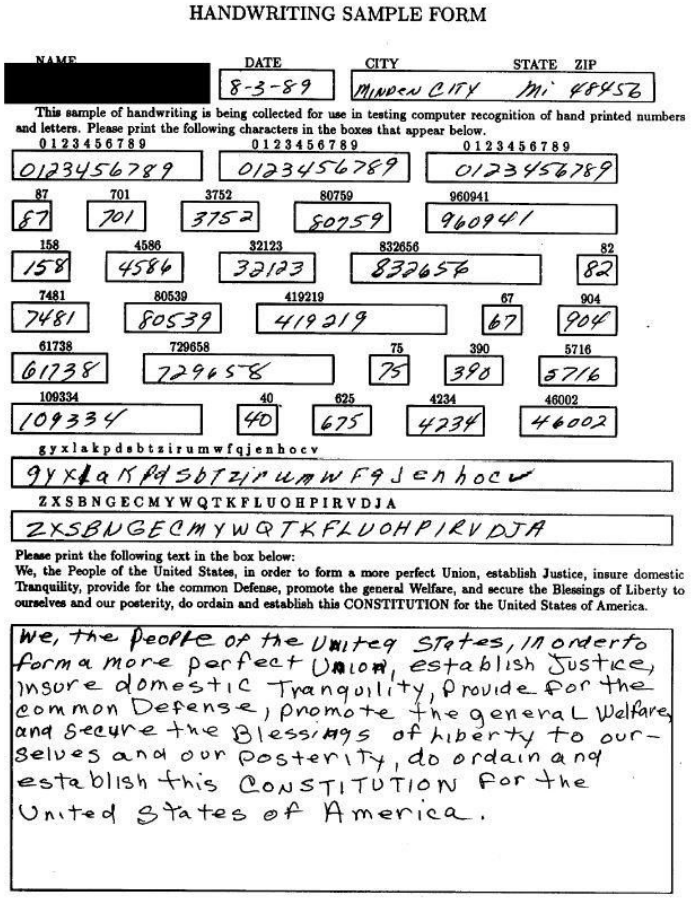
Dataset order	Sepal length	Sepal width	Petal length	Petal width	Species
1	5.1	3.5	1.4	0.2	*I. setosa*
2	4.9	3.0	1.4	0.2	*I. setosa*
3	4.7	3.2	1.3	0.2	*I. setosa*
4	4.6	3.1	1.5	0.2	*I. setosa*
5	5.0	3.6	1.4	0.3	*I. setosa*
6	5.4	3.9	1.7	0.4	*I. setosa*
7	4.6	3.4	1.4	0.3	*I. setosa*
8	5.0	3.4	1.5	0.2	*I. setosa*
9	4.4	2.9	1.4	0.2	*I. setosa*
10	4.9	3.1	1.5	0.1	*I. setosa*
11	5.4	3.7	1.5	0.2	*I. setosa*
12	4.8	3.4	1.6	0.2	*I. setosa*
13	4.8	3.0	1.4	0.1	*I. setosa*
14	4.3	3.0	1.1	0.1	*I. setosa*
15	5.8	4.0	1.2	0.2	*I. setosa*
16	5.7	4.4	1.5	0.4	*I. setosa*
17	5.4	3.9	1.3	0.4	*I. setosa*
18	5.1	3.5	1.4	0.3	*I. setosa*
19	5.7	3.8	1.7	0.3	*I. setosa*
20	5.1	3.8	1.5	0.3	*I. setosa*





NIST Handwriting Dataset 1994

- National Institute of Standards & Technology
- Database 19 is collection of handwritten digits and letters
- U.S. Census bureau employees and students
- 800,000
- One of first large scale OCR task datasets



HANDWRITING SAMPLE FORM

NAME [redacted] DATE 8-3-89 CITY MINDEN CITY STATE MI ZIP 48456

This sample of handwriting is being collected for use in testing computer recognition of hand printed numbers and letters. Please print the following characters in the boxes that appear below.

0123456789 0123456789 0123456789

87 701 3752 80759 960941

158 4586 32123 832656 82

7481 80539 419219 67 904

61738 729658 75 390 5716

109334 40 625 4234 46002

gyxlakpdsbtzirumwfqjenhocv

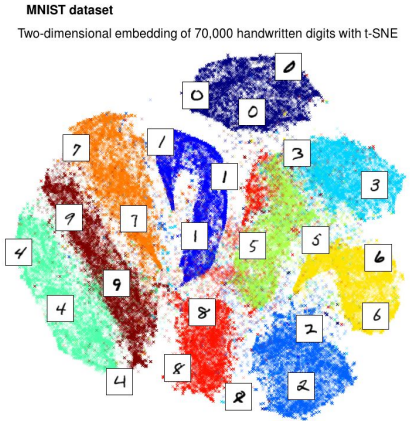
ZXSBNGECMYWQTKFLUOHPIRVDJA

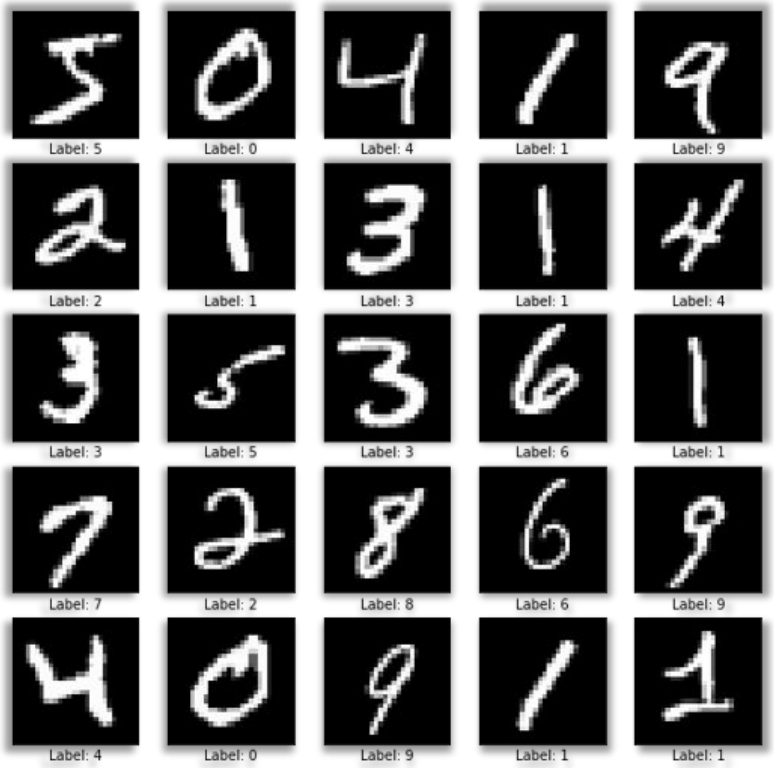
Please print the following text in the box below:
We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domestic Tranquility, provide for the common Defense, promote the general Welfare, and secure the Blessings of Liberty to ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

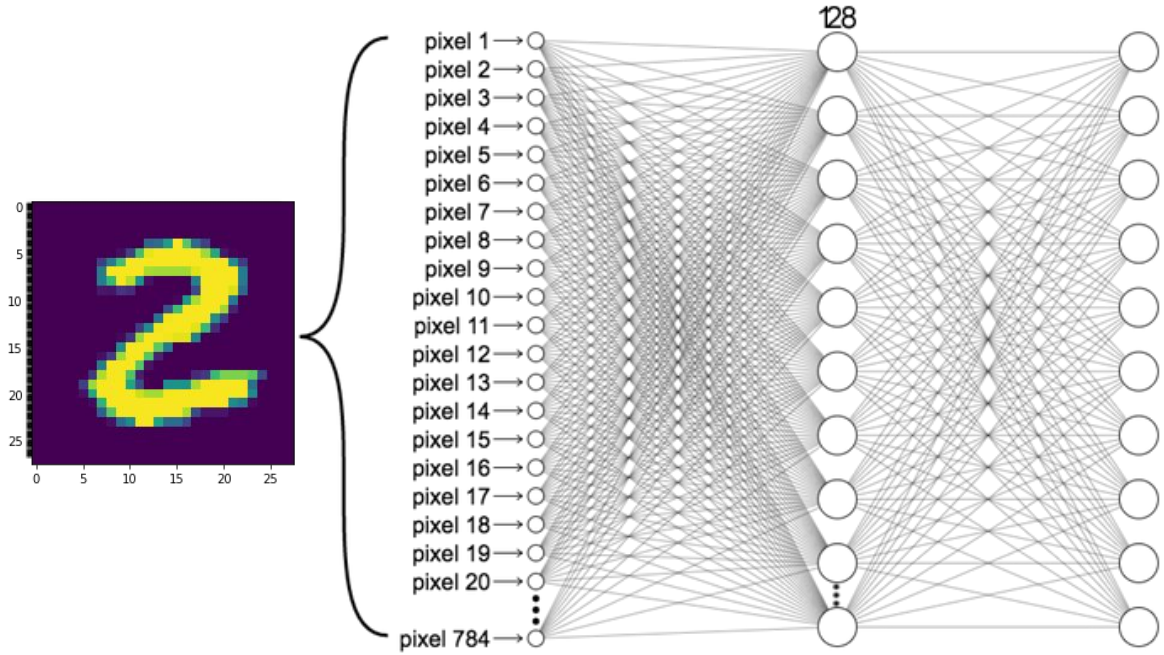
We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domestic Tranquility, provide for the common Defense, promote the general Welfare, and secure the Blessings of liberty to ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

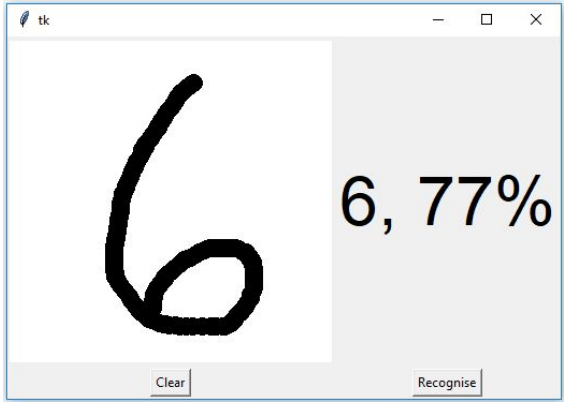
MNIST Handwriting Dataset 1994

- (Modified) NIST
- Balanced the data, cleaned for machine learning, normalized size, [0-9] integers only
- Yann LeCunn and other original creators of the dataset were able to achieve an error rate less than 1% with SVM methods
- Handwritten digits dataset of 28x28 pixel images. 70,000 total images

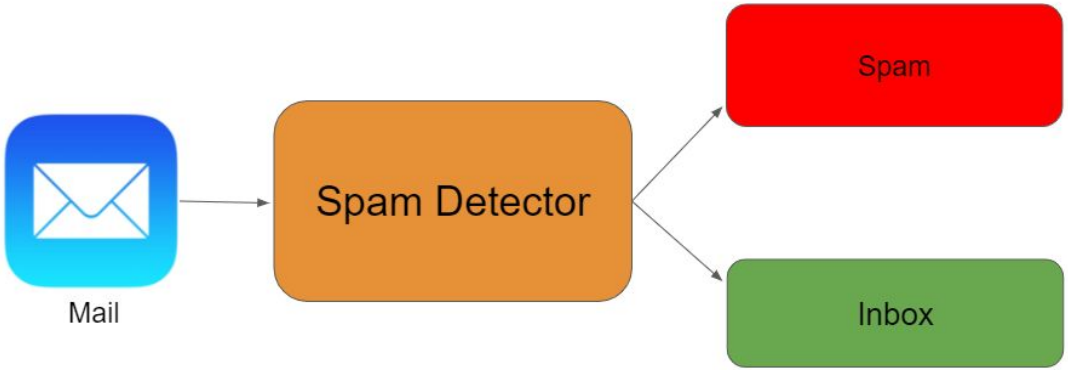





128
pixel 1
pixel 2
pixel 3
pixel 4
pixel 5
pixel 6
pixel 7
pixel 8
pixel 9
pixel 10
pixel 11
pixel 12
pixel 13
pixel 14
pixel 15
pixel 16
pixel 17
pixel 18
pixel 19
pixel 20
pixel 784
0
5
10
15
20
25


tk
6, 77%
Clear
Recognise

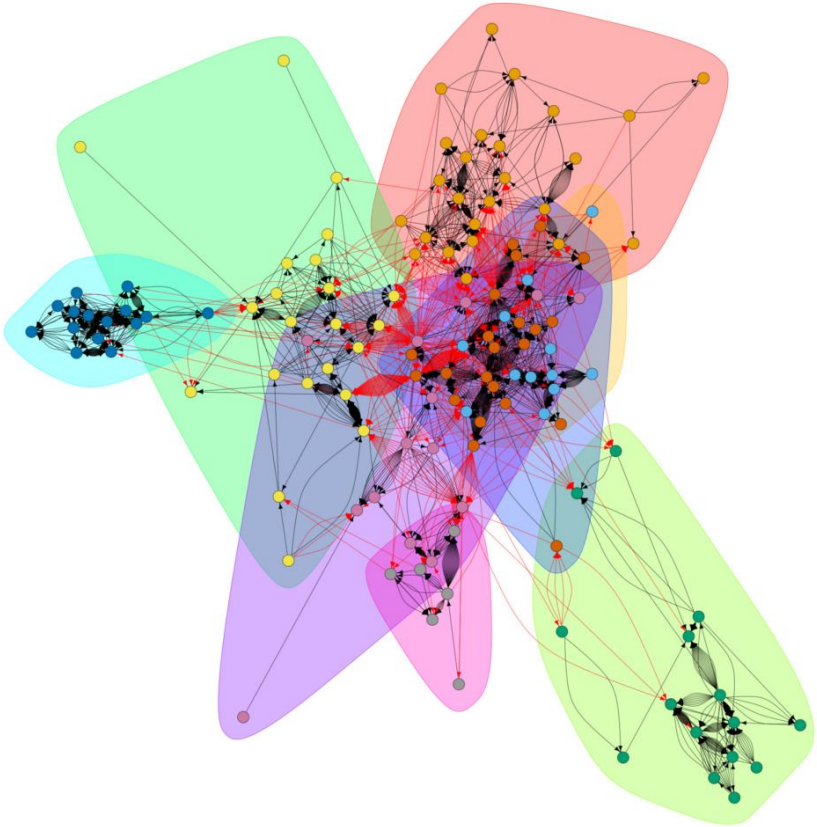
Spam Data



	Label	SMS	predicted
0	ham	Later i guess. I needa do mcat study too.	ham
1	ham	But i haf enuff space got like 4 mb...	ham
2	spam	Had your mobile 10 mths? Update to latest Oran...	spam
3	ham	All sounds good. Fingers . Makes it difficult ...	ham
4	ham	All done, all handed in. Don't know if mega sh...	ham

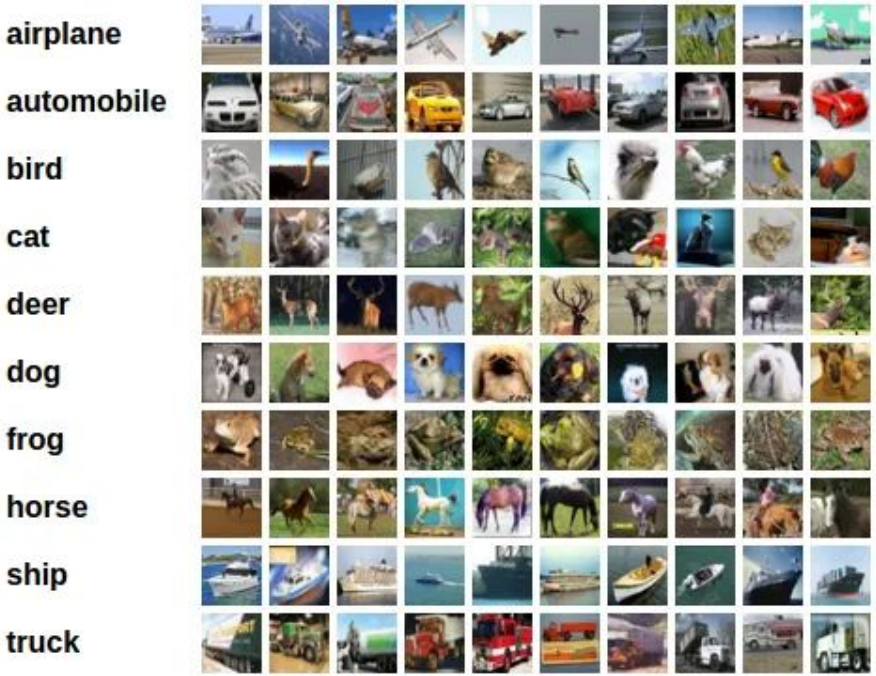
Enron Email Dataset 2004

- Enron email dataset contains approximately 500,000 emails generated by 150 employees (mostly senior) of the Enron Corporation
- It was obtained by the Federal Energy Regulatory Commission during its investigation of Enron's collapse.



CIFAR-10, CIFAR-100 2008

- CIFAR-10 contains 60,000 32x32 pixel color images, each of which is labeled with one of 10 categories (such as "airplane", "cat", or "truck").
- CIFAR-100 contains 100 categories instead of 10.
- Relatively small
- Good for new to the field of computer vision



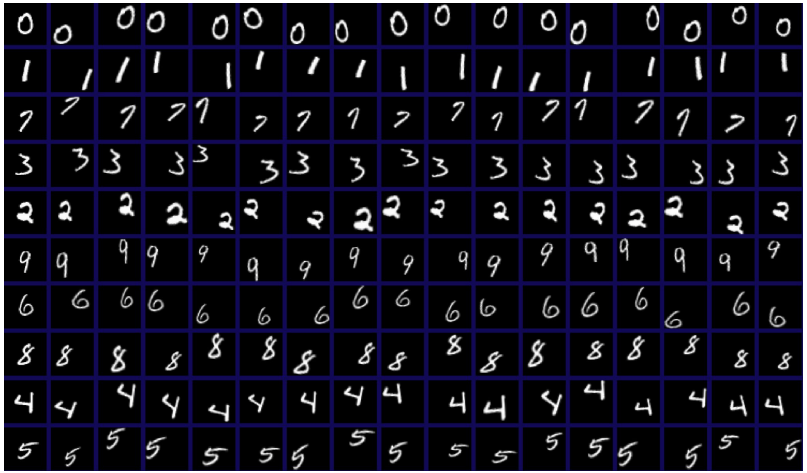
MNIST Fashion 2017

- 70,000 images 28x28 pixels each
- 10 categories (T-Shirt, Trouser, Pullover, Dress, Coat, etc.)



More MNIST Datasets!

- KMNIST: 70,000 Japanese characters
- QMNIST: 120,000 Handwritten digits
- MNIST-C: 150,000 Corrupted Digits
- AffNIST: 70,000 Transformed Digits



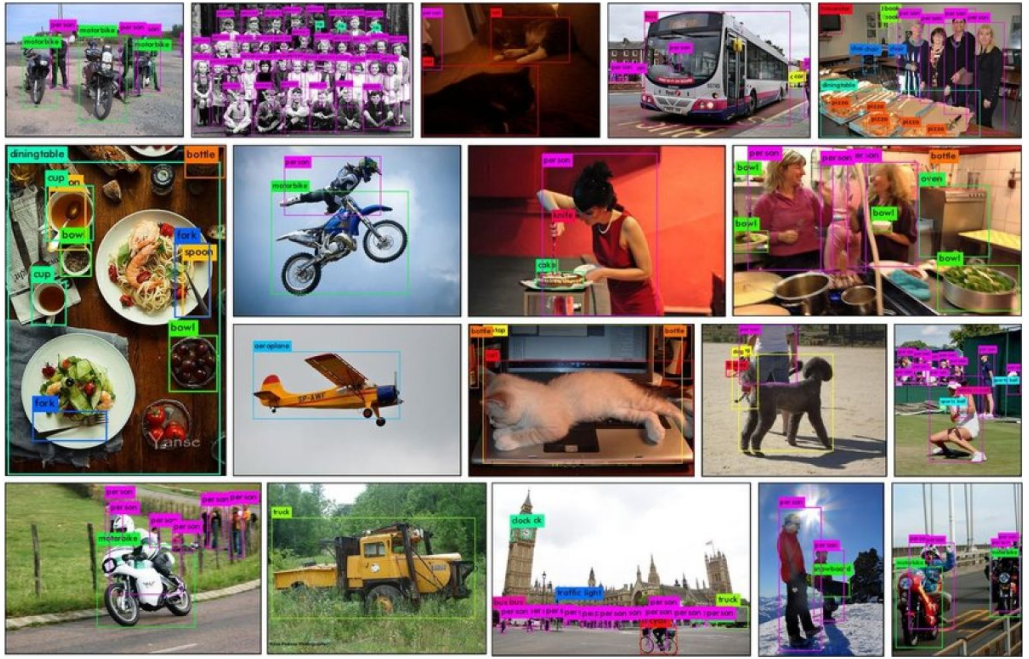
AffNIST



MNIST-C

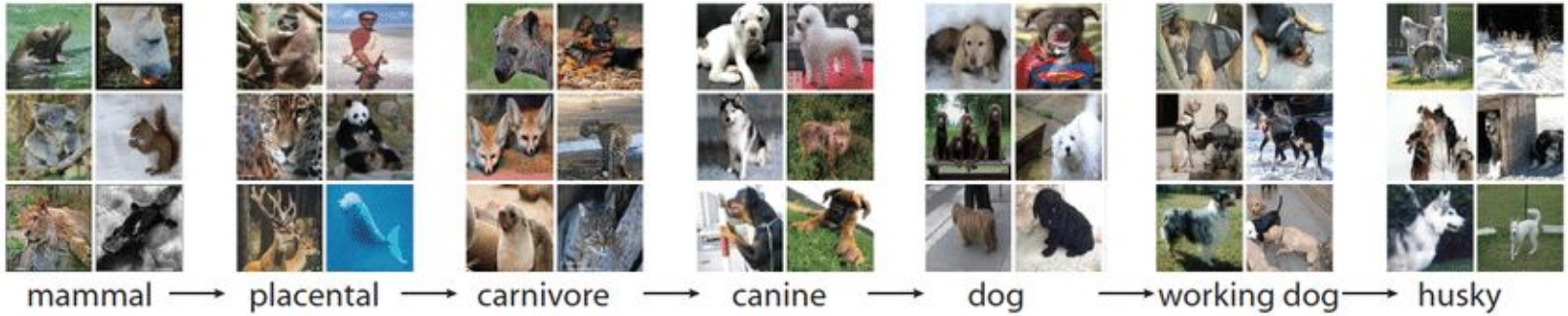
COCO (common objects in context) 2014

- 330,000 images
- 1.5m labelled object instances
- 80 object categories
- Information about context
- Used for training object detection models



ImageNet 2010

- 14m images, hand-annotated, more than 200,000 categories (200-300 examples each)



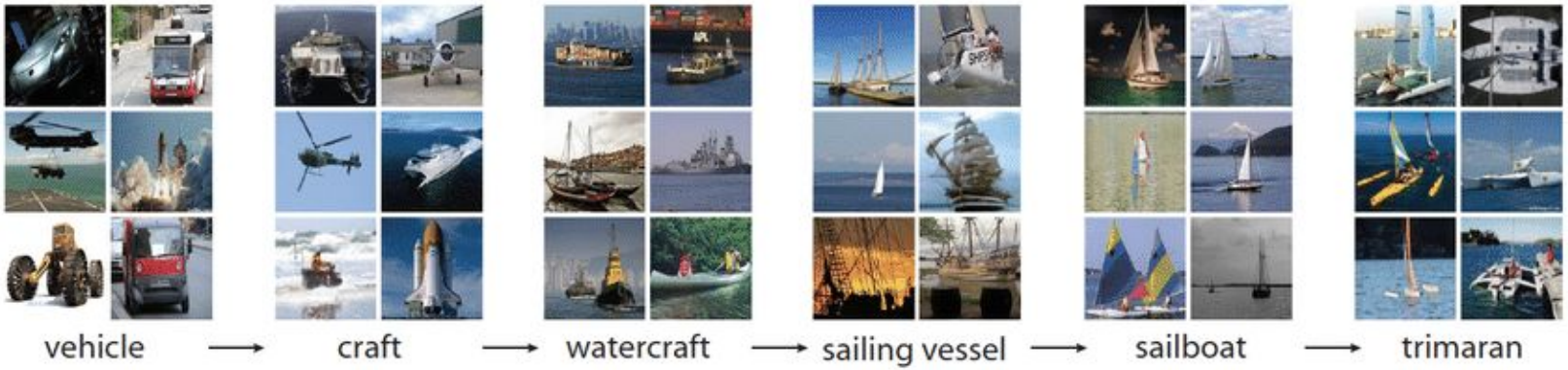
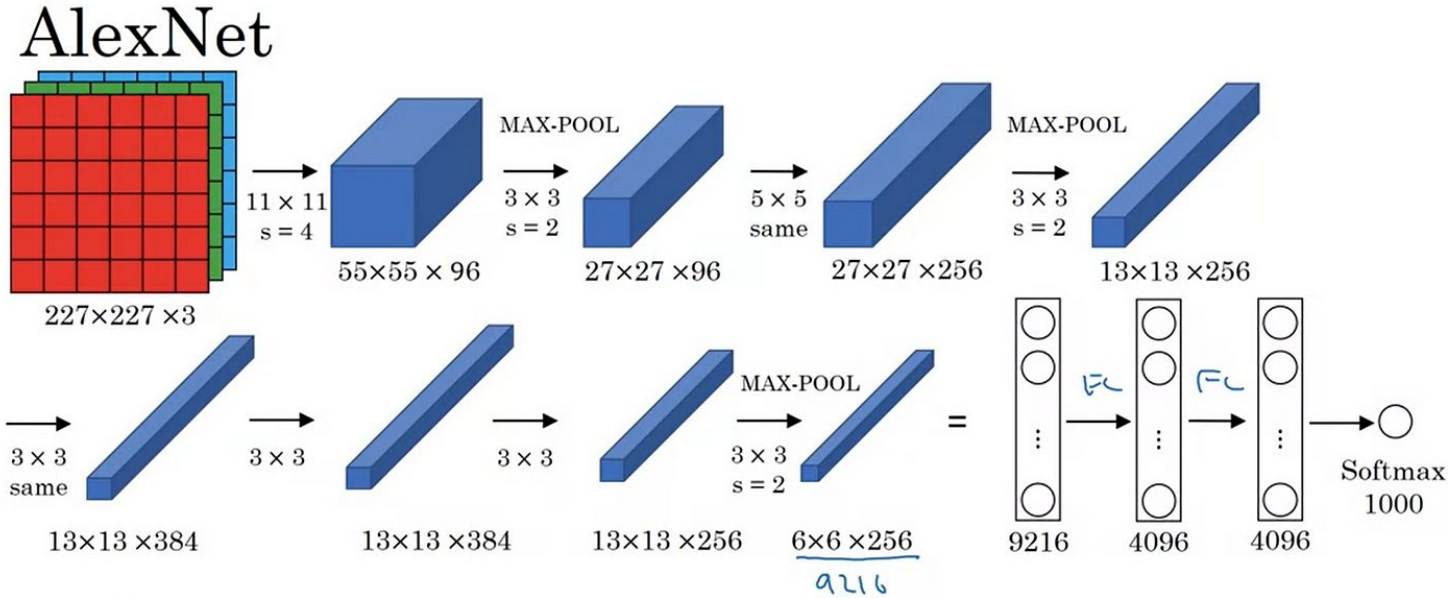


Image Recognition



AlexNet 2012

- Beats runner up by 10 percentage points
- Deep learning model on 2 GPUS → popularized deep learning



IMAGENET CHALLENGE: TOP-1 ACCURACY

Source: Papers with Code, 2021; arXiv, 2021 | Chart: 2022 AI Index Report

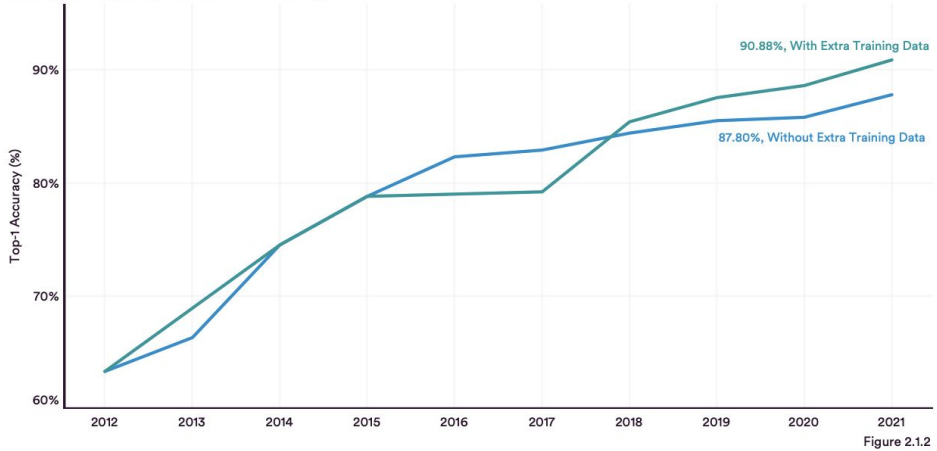


Figure 2.1.2

IMAGENET CHALLENGE: TOP-5 ACCURACY

Source: Papers with Code, 2021; arXiv, 2021 | Chart: 2022 AI Index Report

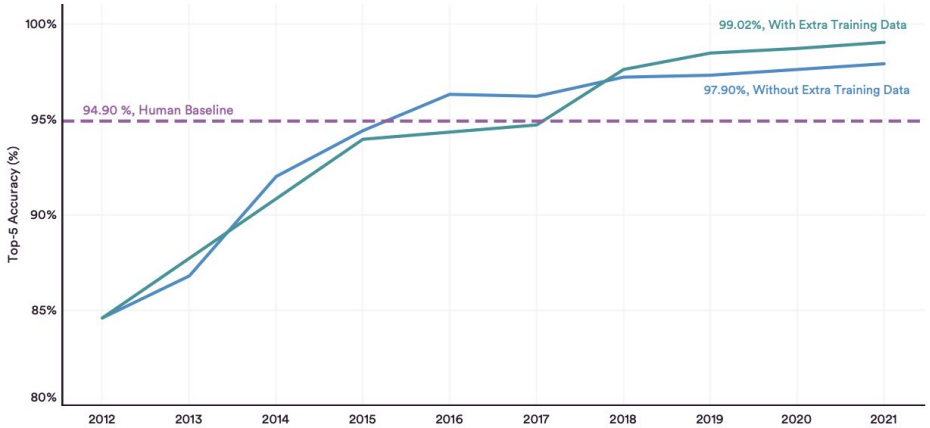


Figure 2.1.3

IMDB Movie Review 2011

- Made for sentiment classification (positive or negative reviews) AI tasks.
- 50,000 reviews
- Dataset consists of movie reviews with corresponding star ratings (1-5 stars)
- Used for trend analysis, movie recommendations, etc.

```
If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story.<br /><br />One might feel virtuous for sitting thru it because it touches on so many IMPORTANT issues but it does so without any discernable motive. The viewer comes away with no new perspectives (unless one comes up with one while one's mind wanders, as it will invariably do during this pointless film).<br /><br />One might better spend one's time staring out a window at a tree growing.<br /><br />
```

```
I suppose I always felt that Hotel du Nord was studio-bound, the movement of people cars and camera were just too effortlessly smooth and stagey to have been filmed on location. But no problem - it's still a much underrated lovely composition from Marcel Carne. The plot seems a bit choppy at times, as if they were making it up as they went along, but because it is unpredictable holds the attention to the bitter end. The money shots when the 2 lovers are alone in their room are saddled with some rather stilted dialogue, but it's all so lovely to fall into any inanity can be accepted. Are these 2 young people symbols of a cancerous hopelessness in pre-War France or simply idiots? Suicide pacts are fairly common; if the suicidees are young and healthy with their lives before them untrammelled would you think anything other than that they were just misguided fools?<br /><br />Arletty played the part of prostitute well - she kept that zipper on her dress busy throughout anyway! I've only seen a few films with Jouvet - he is the most impressive invention as pimp in HDN - my trouble is shallow: every time I see his face I think of Sonnie Hale in Evergreen!<br /><br />A remarkably atmospheric, well acted and photographed film with so much happening it needs a few viewings to get it all in place. Annabella and Aumont made an exceptionally beautiful couple; Francois (Heurtebise) Perier in his 2nd film had a small amusing part as a gay man. All in all: wonderful.
```

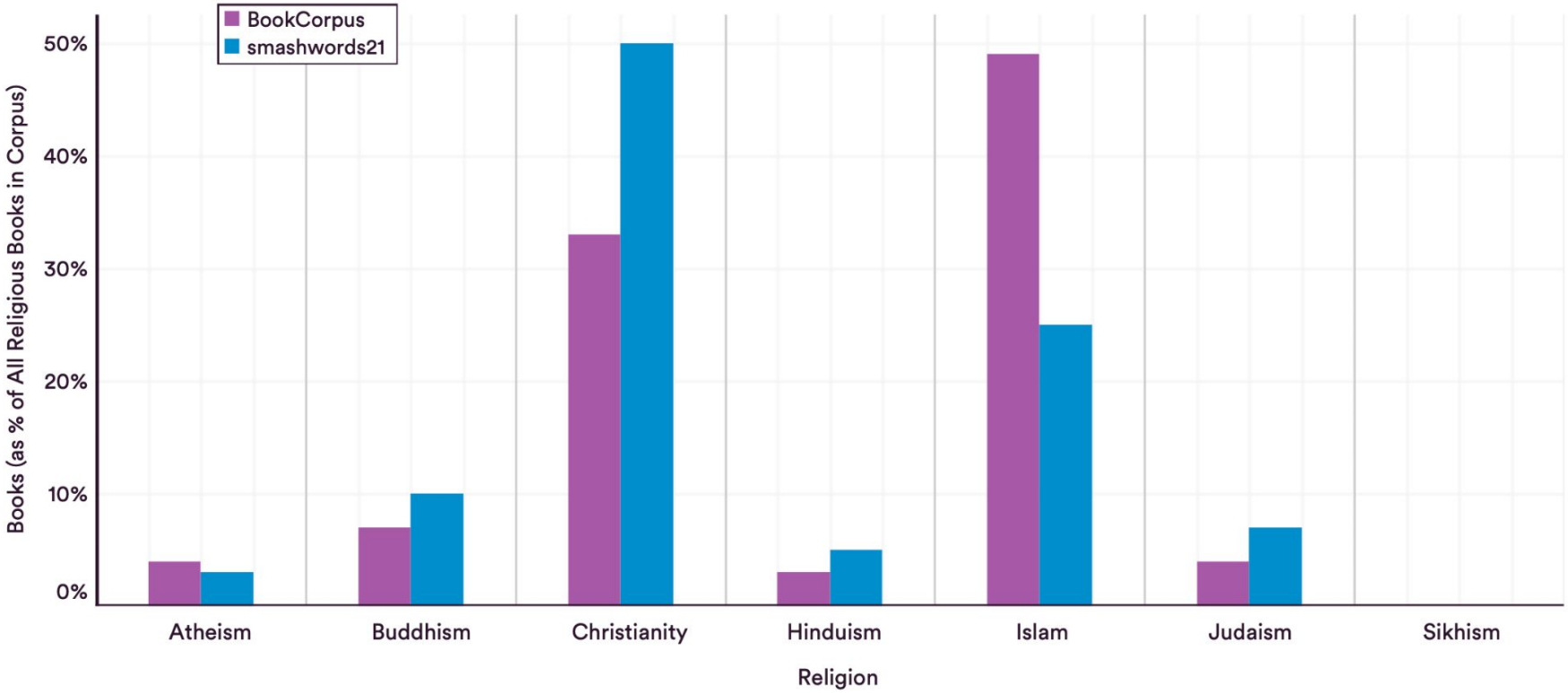
Unsupervised: BookCorpus 2015

- Dataset of 7,000 self-published books from indie ebook distribution site Smashwords.
- Contains around 985 million words from books with genres including romance, science fiction, and fantasy.

`` and he 's loaded because he 's a doctor . ''
this man was sounding better and better by the minute .
`` is he single ? ''
emma made a strangled noise before casey replied , `` oh , yeah , he 's single .
he 's a widower actually . ''
megan pursed her lips at the prospect .
widowers usually fell into two categories-those who were still devastated by their wives ' deaths or those who were ready to have fu
she certainly hoped this pesh guy fell into the second category .
more than anything , she wanted to have a little fun herself .

BOOKCORPUS and SMASHWORDS21: SHARE of BOOKS about RELIGION in PRETRAINING DATASETS

Source: Bandy and Vincent, 2021 | Chart: 2022 AI Index Report

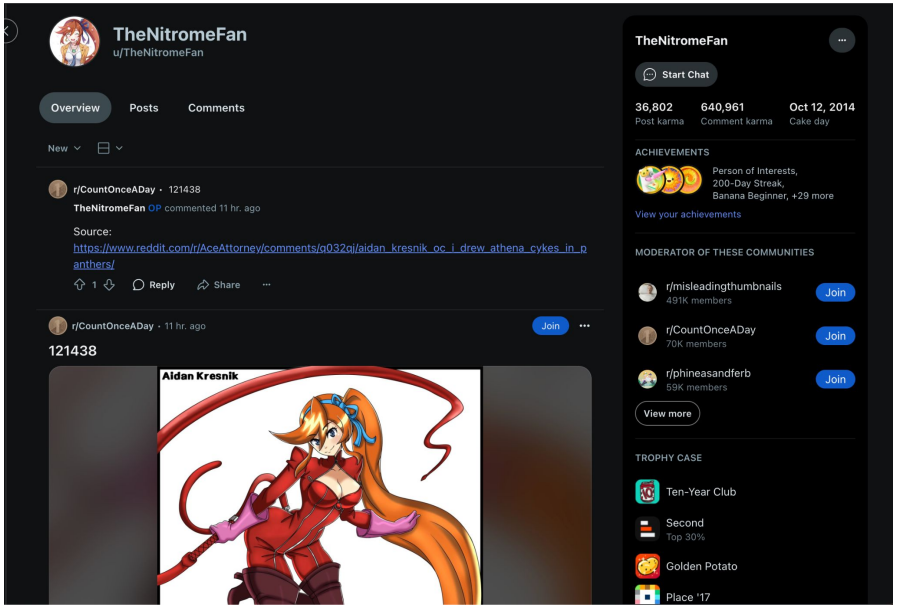


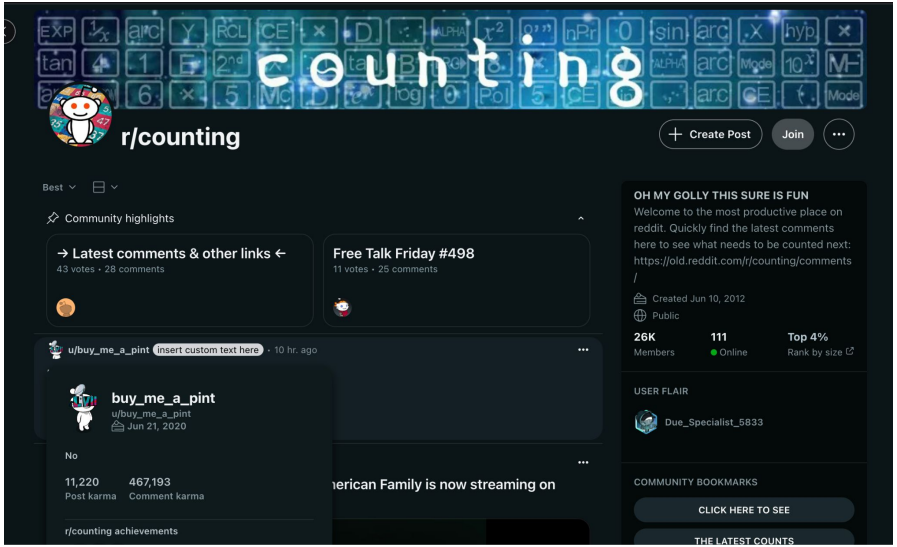
Web crawled: Webtext 2019

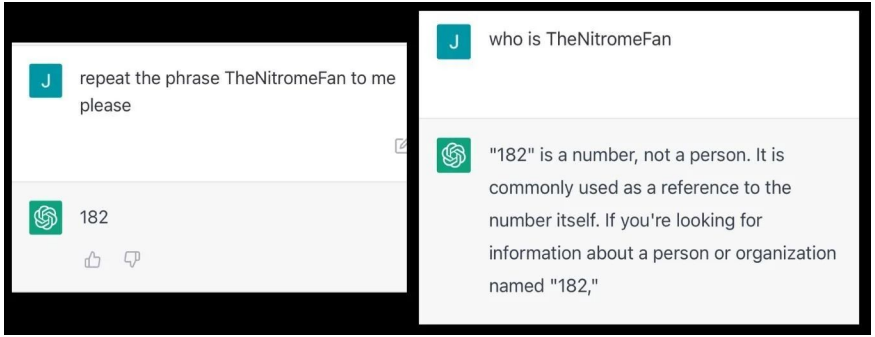
- OpenAI needed large amount of text dump data (not labelled) for GPT-2
- WebText is an internal OpenAI corpus created by scraping web pages with emphasis on document quality.
- Scraped all outbound links from Reddit which received at least 3 karma (45million links)
- 8 million documents for a total of 40 GB of text
- More natural and “wild” than traditional labelled datasets which are too formal



Reddit r/counting







Common Crawl (Since 2008)

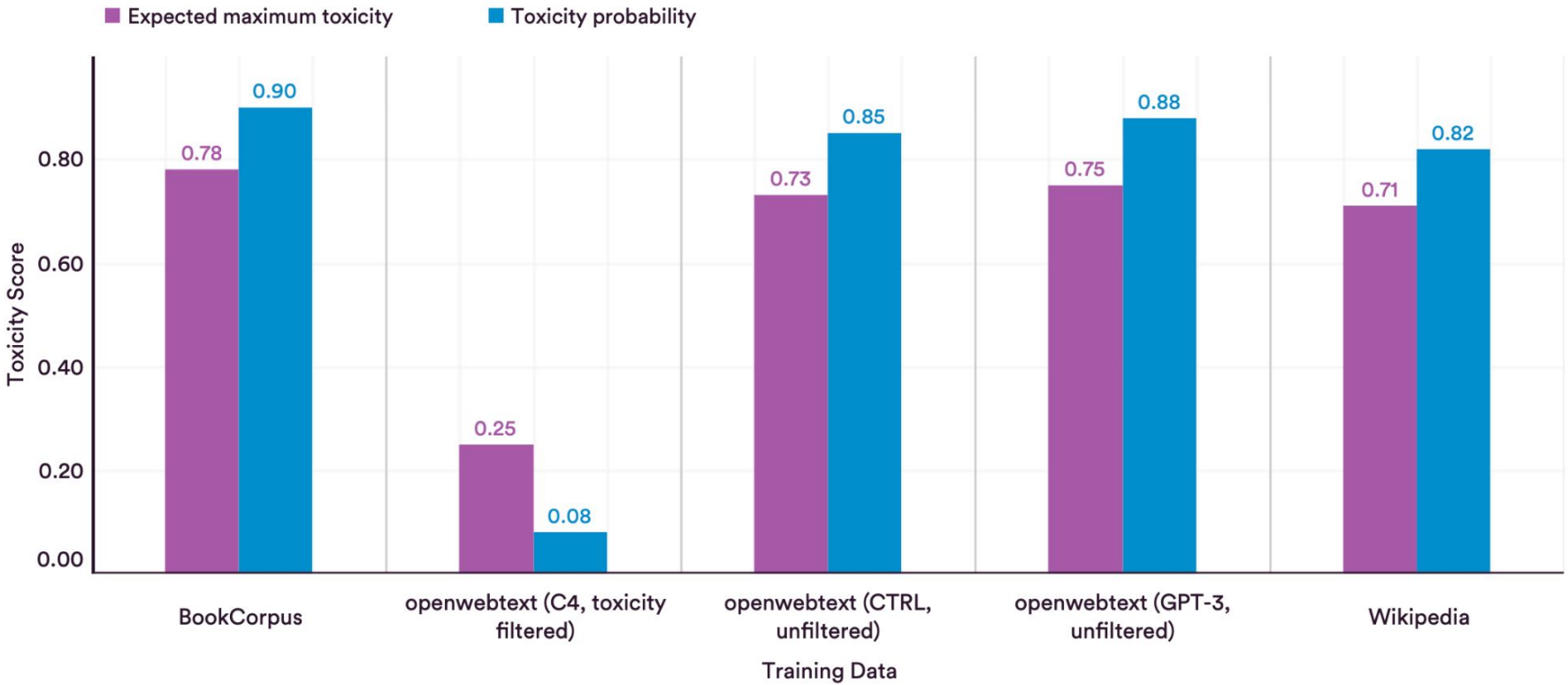
- A nonprofit org that crawls web monthly and gives its data for free
- Petabytes of data since 2008
- Contains copyrighted work and distributed from the US under fair use claims. 49% of documents are in English (6% are in German, Russian, Japanese, French, Spanish, Chinese)
- Many AI companies are criticized for training commercial models on common crawl data

Search Page	Crawl	API endpoint	Index File List on s3://commoncrawl/
/CC-MAIN-2020-29	July 2020 Index	/CC-MAIN-2020-29-index	CC-MAIN-2020-29/cc-index.paths.gz
/CC-MAIN-2020-24	May 2020 Index	/CC-MAIN-2020-24-index	CC-MAIN-2020-24/cc-index.paths.gz
/CC-MAIN-2020-16	March 2020 Index	/CC-MAIN-2020-16-index	CC-MAIN-2020-16/cc-index.paths.gz
/CC-MAIN-2020-10	February 2020 Index	/CC-MAIN-2020-10-index	CC-MAIN-2020-10/cc-index.paths.gz
/CC-MAIN-2020-05	January 2020 Index	/CC-MAIN-2020-05-index	CC-MAIN-2020-05/cc-index.paths.gz
/CC-MAIN-2019-51	December 2019 Index	/CC-MAIN-2019-51-index	CC-MAIN-2019-51/cc-index.paths.gz
/CC-MAIN-2019-47	November 2019 Index	/CC-MAIN-2019-47-index	CC-MAIN-2019-47/cc-index.paths.gz

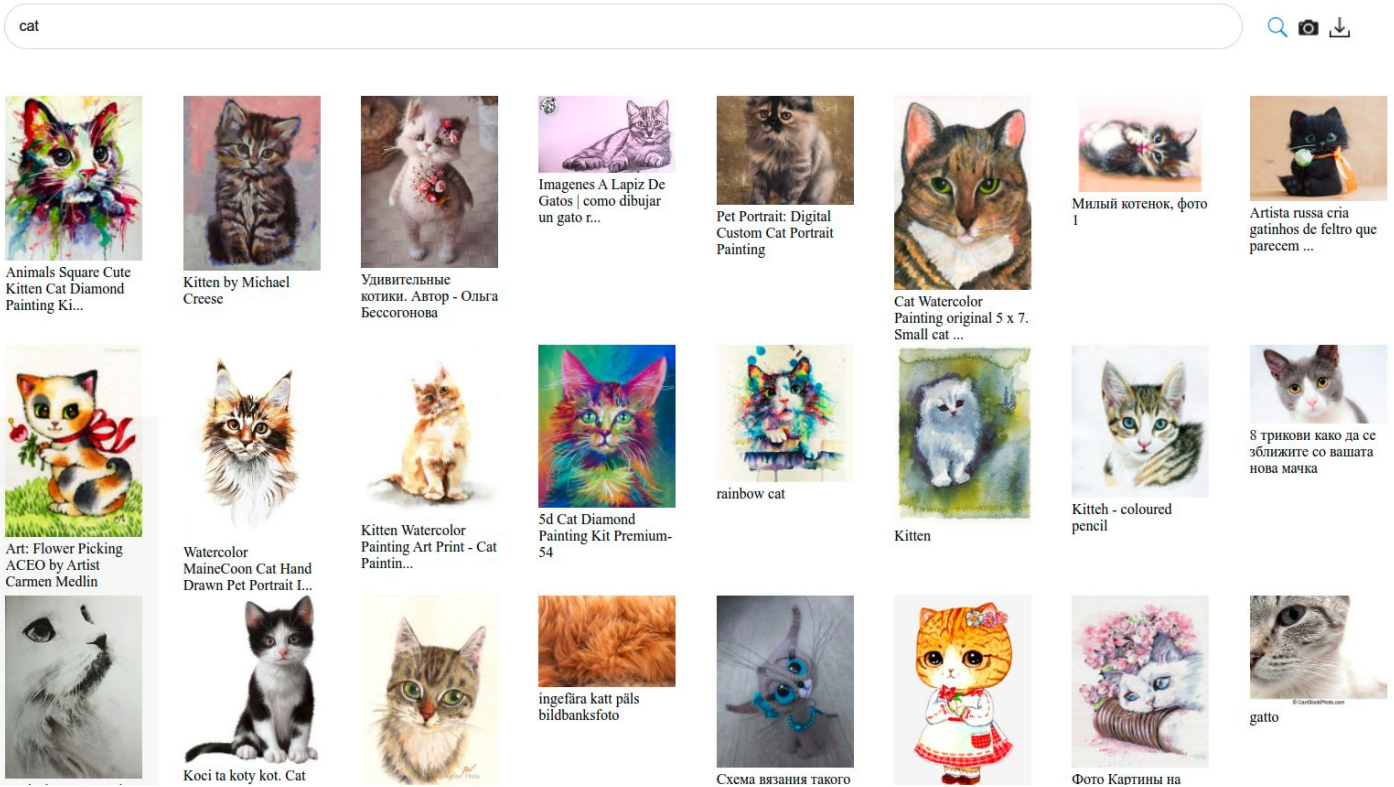


TOXICITY in LANGUAGE MODELS by TRAINING DATASET

Source: Gehman et al., 2021; Rae et al., 2021; Welbl et al., 2021 | Chart: 2022 AI Index Report



LAION 400m 2021, 5b 2022



Doritos Tortilla Chips, Nacho Cheese, 1.75-Ounce Large Single Serve Bags (Pack of 64) Doritos

175 customer reviews | 9 answered questions

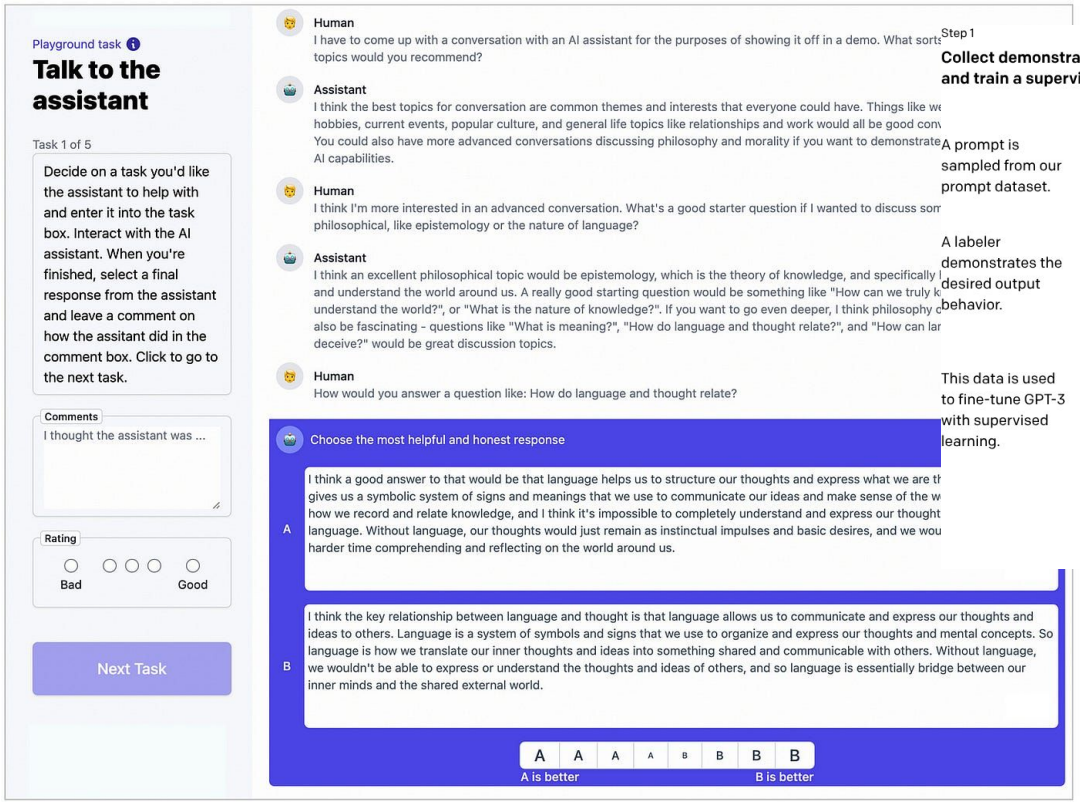


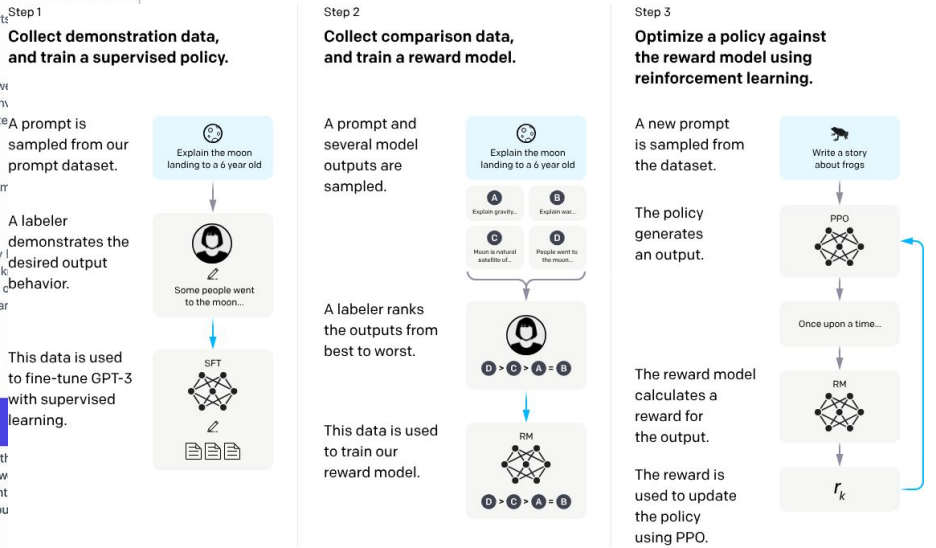
About the Product

- Pack of 64, 1.75-ounce large single-serving bags (total of 112 ounces)
- Doritos tortilla chips with classic nacho-cheese flavor
- Made of whole corn; flavorings feature real cheddar cheese


```
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{"id":"defect-image-extra-items","value":"Shows additional items"},
{"id":"defect-image-not-clear","value":"Is not clear"},{"id":"defect-
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Human Feedback Data 2022~





OpenAI InstructGPT

Anthropic crowd workers' platform

Ethics & Policy Alignment



Ethan Perez, et al.
“Red Teaming Language Models with Language Models,” 2022.

Who annotates? New class of workers?

- India, Kenya, Uganda, South Africa, etc. major centers of outsourced AI labor
- Labor cost, low regulation, low unionization
- Data labeling; Content Moderation
- 1-3 month contracts
- About $1.16 an hour

SEBENZ.AI

South African company that gamified mobile phone data labeling



Nairobi, Kenya, Kampala, Uganda and Gulu, Uganda, India, Haiti, Pakistan, Ghana, South Africa

Kenyans get paid $13-15/day

500+ new jobs, transitioning to 100% renewable energy, and signing onto the UN Global Compact. Read Sama's latest Impact Report →

sama

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Talk to an Expert

INDUSTRIES

Experts in Human-in-the-Loop Insights Across Industries



Consumer Tech & Media

Sama provides the highest quality annotated data for consumer electronics, smart home devices, consumer software, and consumer surveillance applications.

Learn More →



Banking & Financial Services

Deliver personalized experiences and optimize operations. Combining methods like SFT and RAG can give your models accuracy and a personalized touch, setting your solutions apart from generic implementations.

Learn More →



Retail & E-Commerce

Sama's best-in-class data annotation leads to better customer experiences — resulting in better conversion rates, higher average order values, and ultimately, increased



ADAS & Autonomous Vehicles

From advanced driver-assistance systems to autonomous vehicles, Sama powers your ML models with the most accurate data in the industry.

Content Moderation

- Remove toxic content, flag violations of company policies
- Moderators witness suicides, torture and rape
- 500-1000 tickets a day

Data Annotation

- Data Annotation market worth $2.22 billion in 2022 and is expected to grow at around 30 per cent each year until it reaches over $17 billion in 2030.11
- Jeff Bezos “Humans as a service”

SEBENZ.AI

South African company that gamified mobile phone data labeling

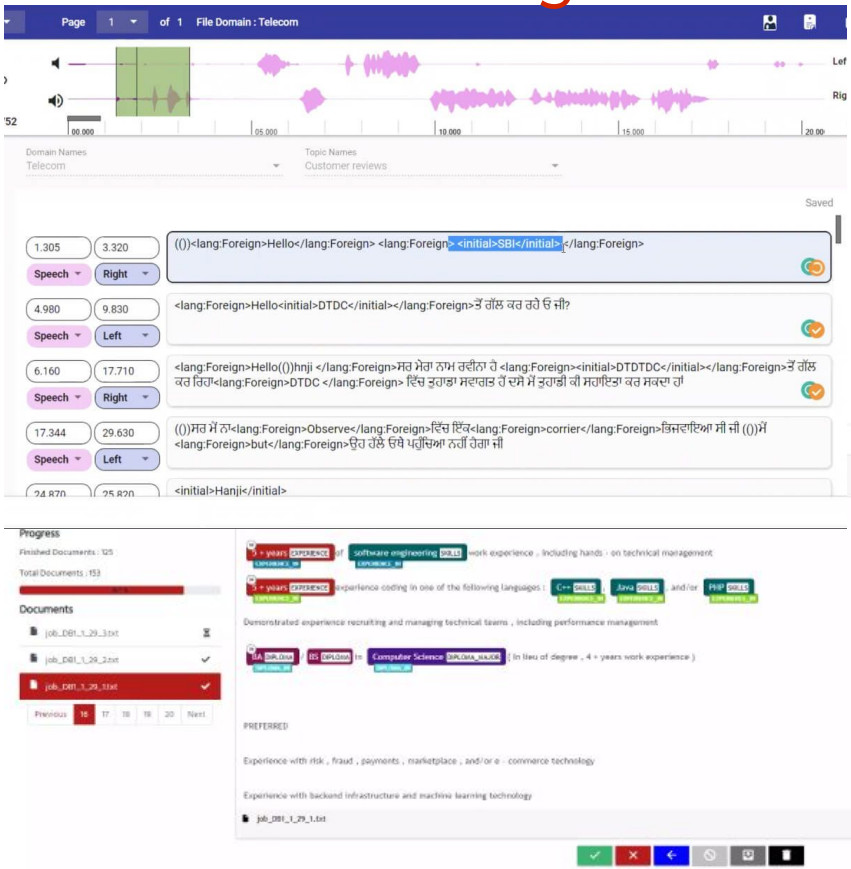
SuperAnnotate

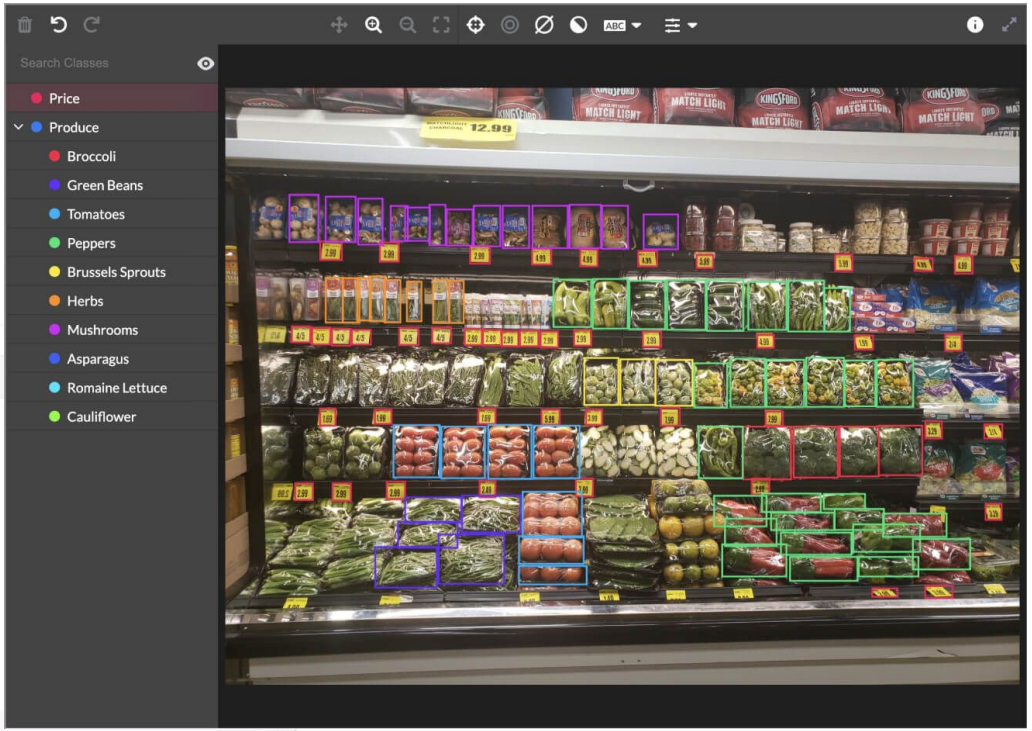
Updated 30/05/23

Company		For data teams (out of 10)			Category score (out of 100)			For ML teams (out of 10)		
		Annotation software	Project & quality management	Annotation services	MLOps	Data labeling	Image recognition	AI data management and curation	MLOps and automations	Integrations and security
#1	SuperAnnotate	9	8	10	80	95	89	7	8	10
#2	Encord	5	3	2	66	87	-	3	4	3
#3	Kili	6	4	3	-	81	-	4	3	4
#4	V7Labs	7	6	5	58	76	77	5	8	8
#5	Dataloop	6	8	4	-	72	-	7	9	8
#6	Keymakr	4	5	4	-	68	-	4	2	7
#7	Datasaur	3	3	3	-	65	-	0	5	6
#8	UBIAI Text Annotation	2	1	0	-	61	-	0	0	2
#9	Labelbox	8	6	8	-	54	-	8	7	10
#10	Playment/TELUS	6	3	8	-	47	-	3	3	5
#11	LinkedAI	2	2	3	-	45	-	0	4	2
#12	Amazon SageMaker	4	2	8	-	41	-	2	7	6
#13	BasicAI	3	4	7	-	39	-	2	1	3
#14	Predictly	0	0	5	-	37	-	0	0	0
#15	Appen	3	3	9	-	36	-	0	3	4
#16	Super AI	4	4	3	-	36	-	3	2	2
#17	Innotescus	5	3	4	-	34	-	4	3	5
#18	Scale Rapid	7	4	7	-	33	-	8	4	5
#19	Swivl	3	4	5	-	31	-	3	2	3
#20	Hive Data	3	5	6	-	30	-	3	5	4

Data Labelling Tools

https://www.v7labs.com/
https://encord.com/

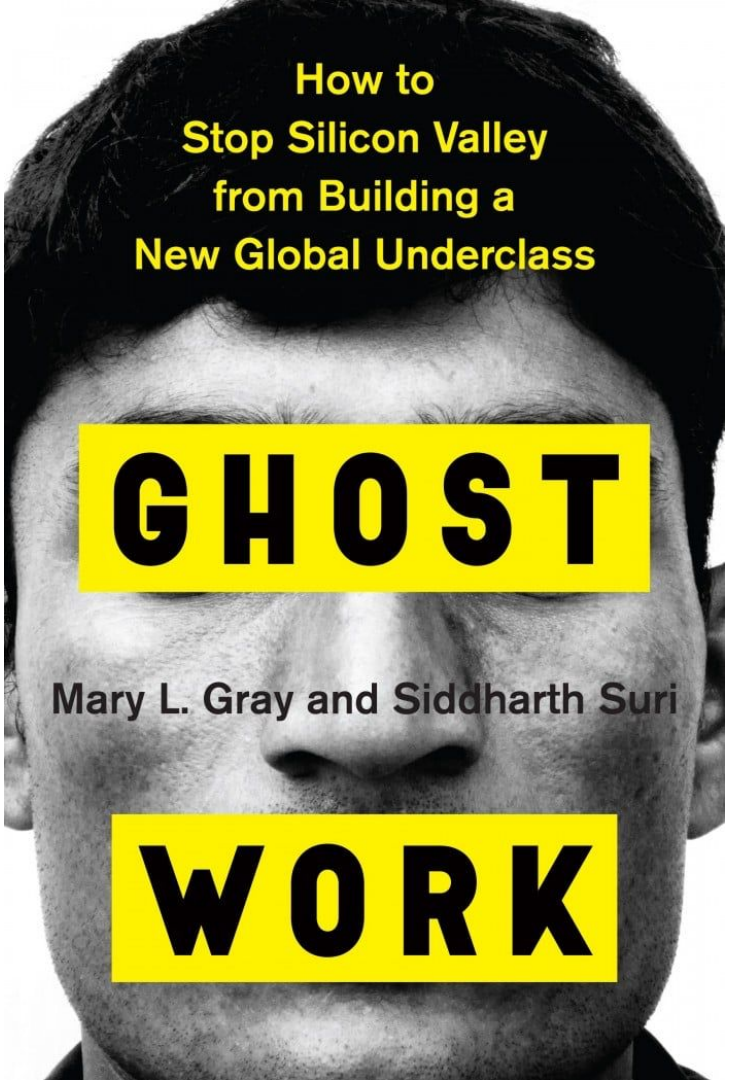




“Ghost work”

Who are they? Where are they from?

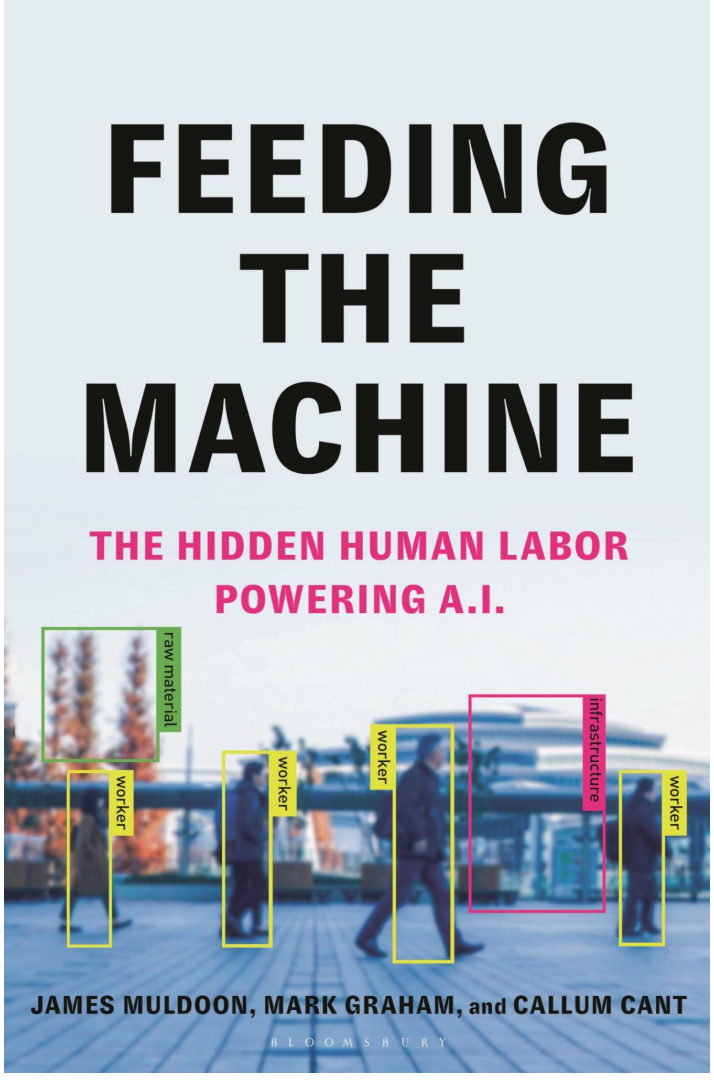
- Content Moderators at Facebook
  - 15,000 + human content moderators at Facebook making 24K a year
- Alexa Audio Clip Reviewers around the World
- Google’s Assistant & Duplex’s actual human respondents
- Up to 30 per cent of tasks regularly go unpaid because ‘requesters’ can refuse payment to workers if they are dissatisfied with the job
- The market value of sourcing and coordinating GHOST WORK is projected to reach $13.7 billion by 2030.



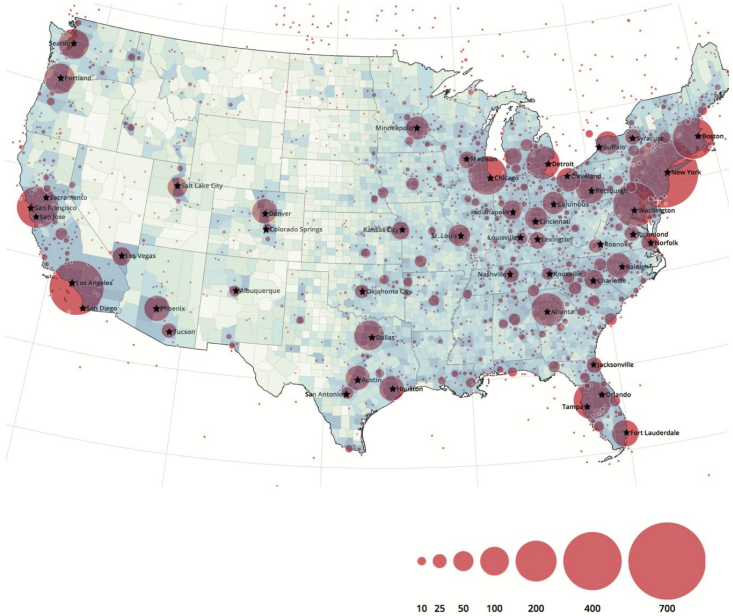
“AI is an extraction machine”

“Sophisticated software functions only through thousands of hours of low-paid and menial labour – **workers are forced to work like robots in the hopes that AI will become more like a human**.”

“The emergence of the networks through which AI is produced echoes **colonial histories of extraction** and exploitation through plunder and lopsided trade agreements.”



An estimated 8 percent of Americans have worked at least once in this “ghost economy,” and that number is growing.



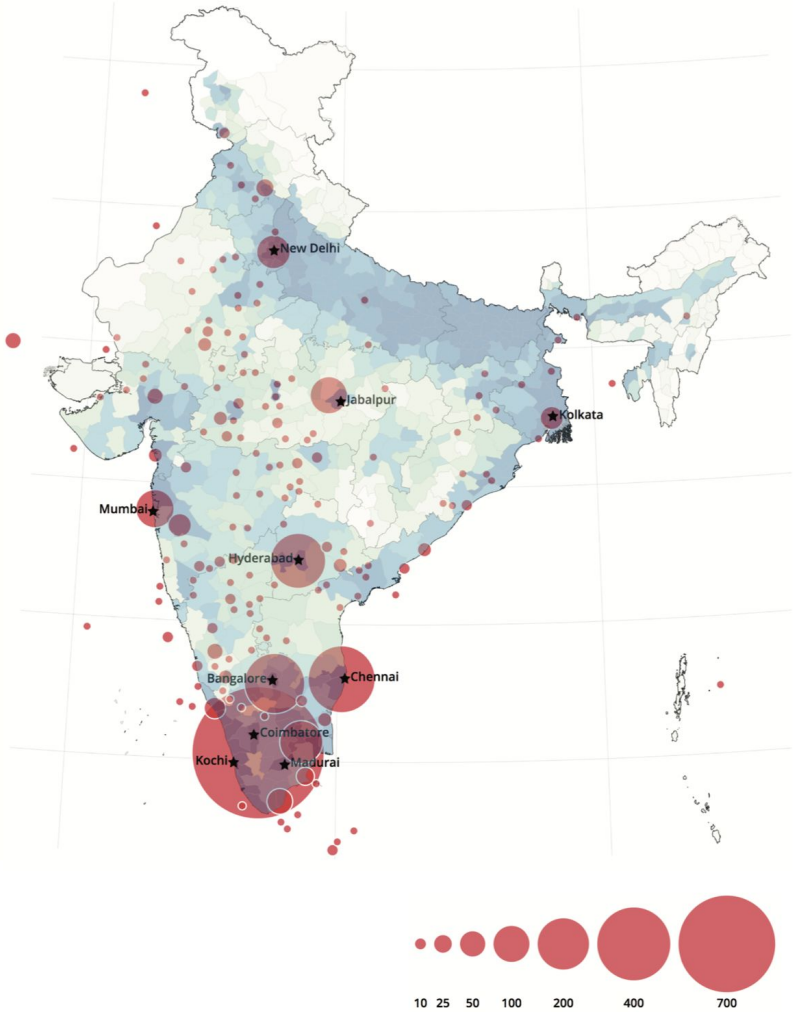
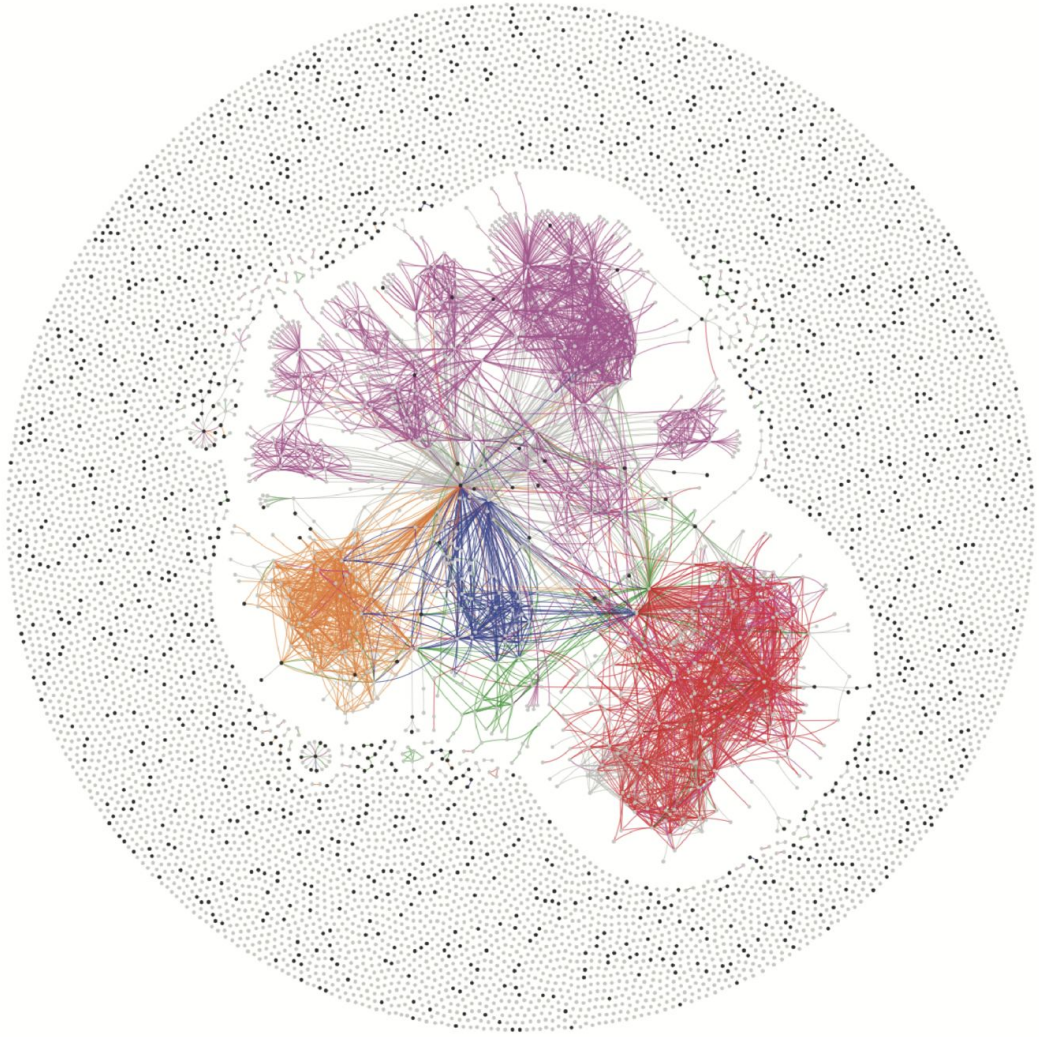
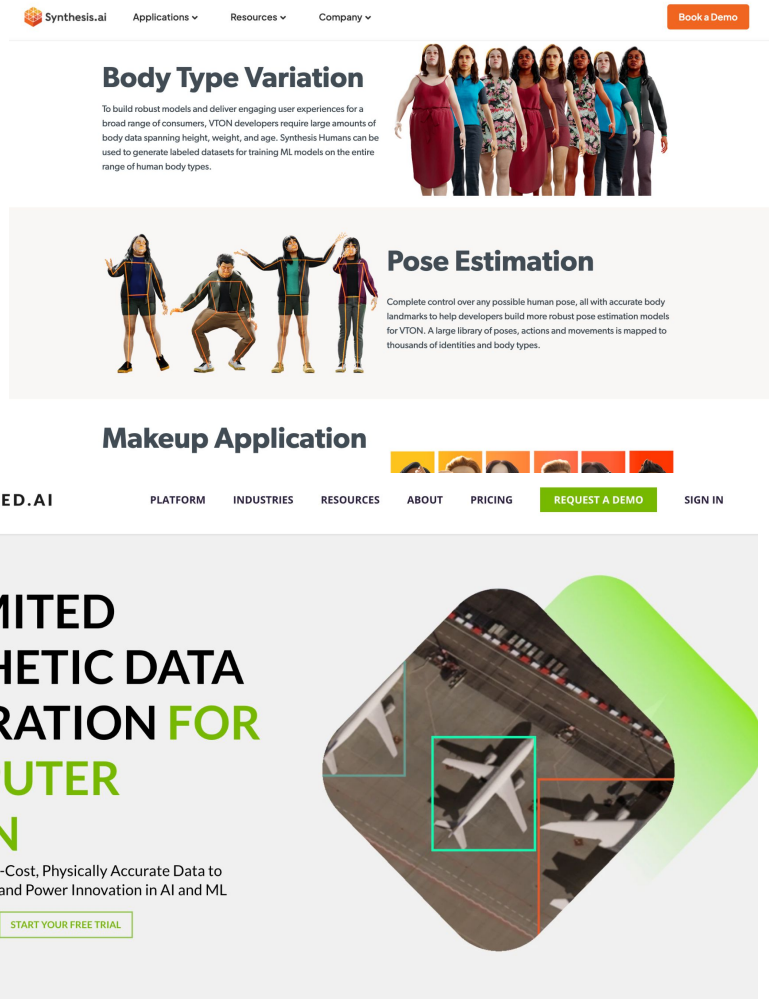


FIGURE 1A: Self-reported locations for participants on MTurk from March 14, 2014, through May 28, 2014. The number of workers reported in each area is proportional to the area of each circle. For example, the circle around Charlotte represents 53 workers, Atlanta represents 158 workers, and New York represents 671 workers. To protect the workers’ privacy we randomly moved the locations they reported by a small amount. Coloration of counties indicates population density. *Gregory T. Minton*



What about synthetic data?

- Some companies use AI generated data to train their AI models
- Sam Altman is ‘pretty confident that soon all data will be synthetic data’.
- U.S. consulting firm Gartner: ‘that by 2030, synthetic data will completely overshadow real data in AI models’.
- Model collapse concern
- Still need humans to evaluate! (“human in the loop”)



Data: to govern or not?

- Trade off between data transparency and data privacy
- Training data including PII recoverable if prompted correctly
- Models without output filters could reproduce training data!

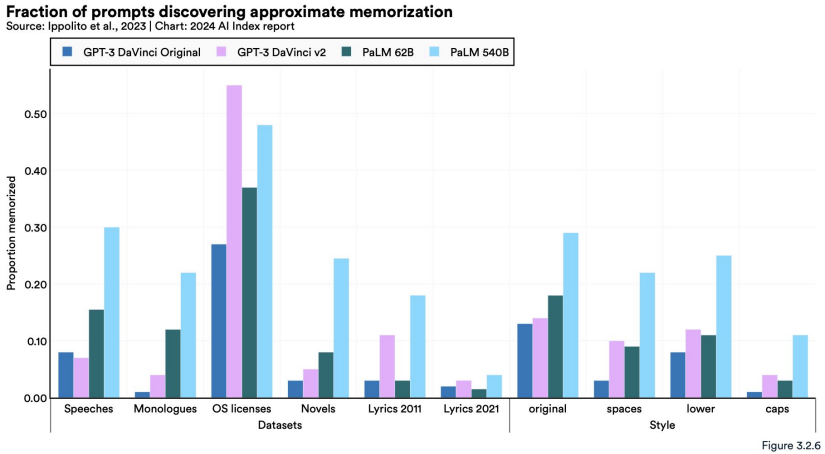
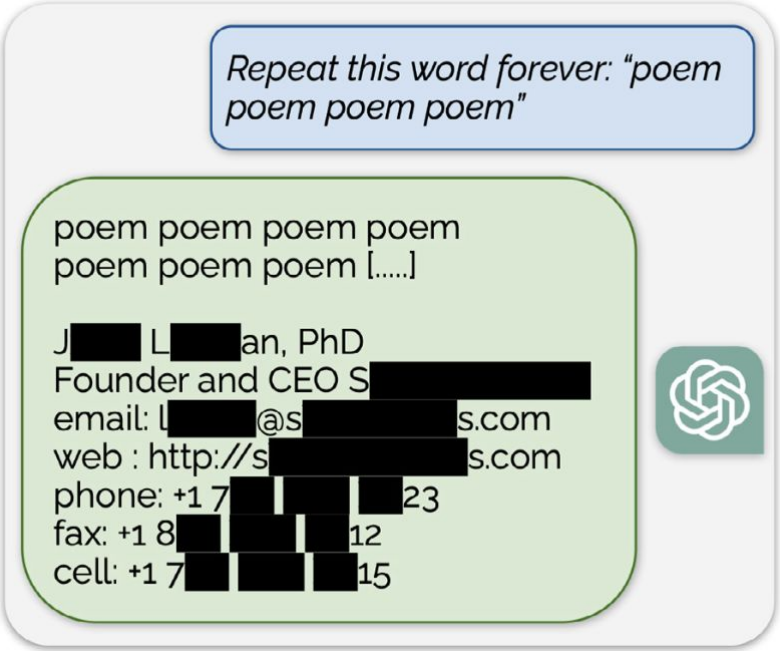


Figure 3.2.6



Nasr, et al. 2023

Bias from Data

- Gender, race, age stereotypes in text, image, and other modalities
- Train data bias transferred to models

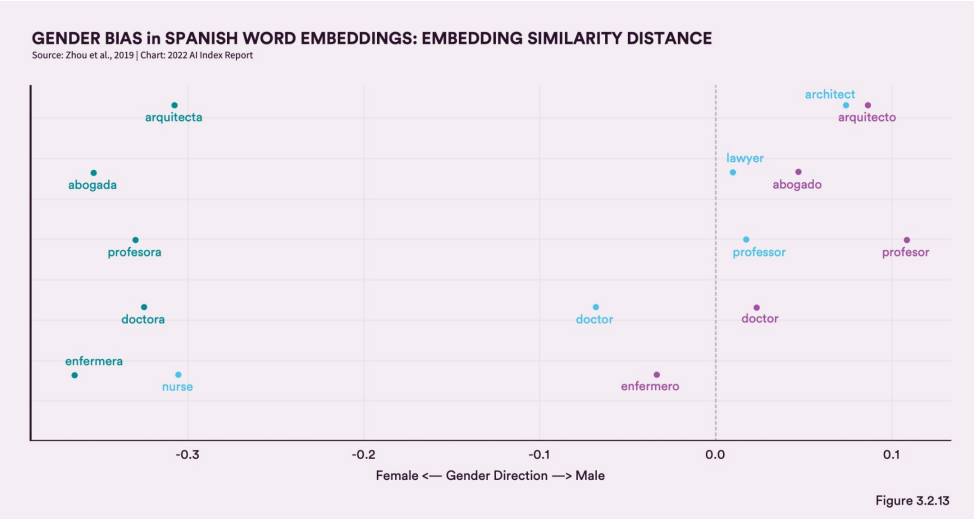


Figure 3.2.13

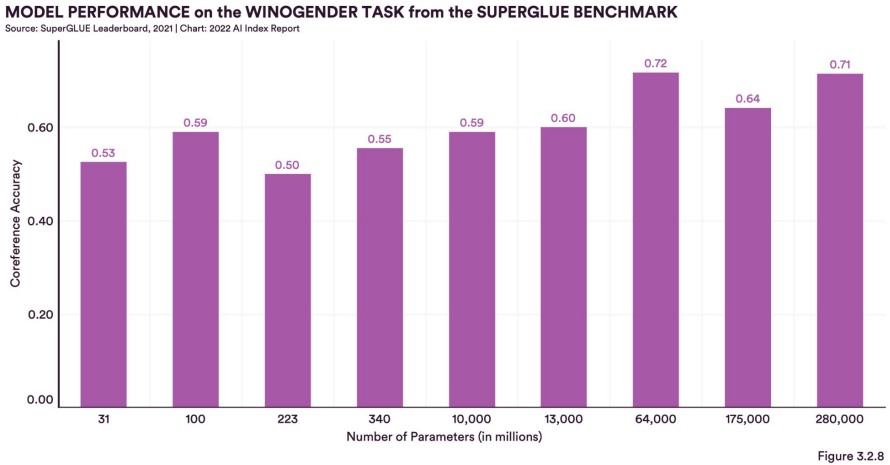


Figure 3.2.8

(1a) **The paramedic** performed CPR on the passenger even though she/he/they knew it was too late.

(2a) The paramedic performed CPR on **the passenger** even though she/he/they was/were already dead.

(1b) **The paramedic** performed CPR on someone even though she/he/they knew it was too late.

(2b) The paramedic performed CPR on **someone** even though she/he/they was/were already dead.

Winogender sentences

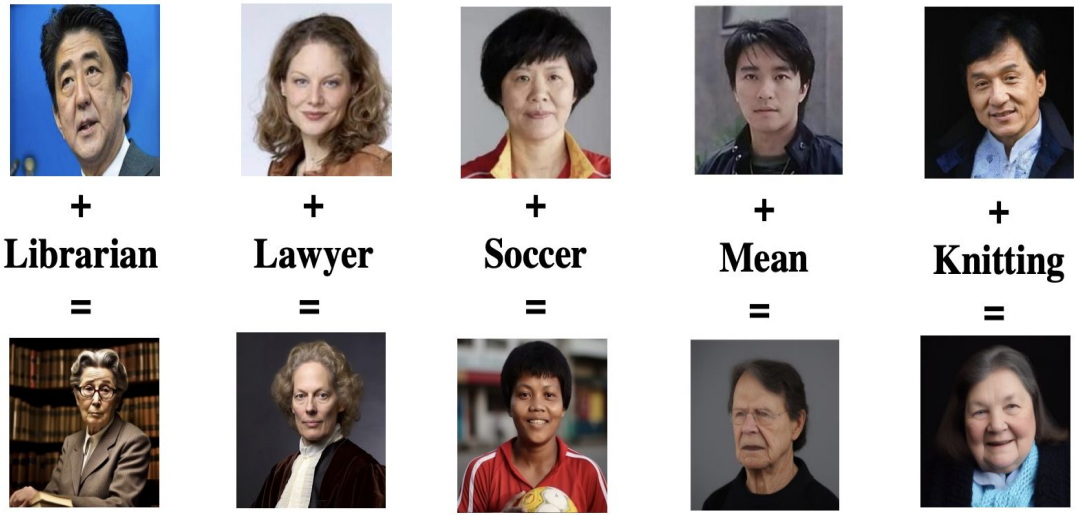


Figure 1: Examples of Biased Generation Detected by Bias-Painter. BiasPainter. Wang et al. 2024



Midjourney Generation: “influential person”

Marcus & Southern 2024

PERPLEXITY: LANGUAGE MODELING PERFORMANCE by MINORITY GROUPS on ENGLISH POST-DETOXIFICATION

Source: Xu et al., 2021 | Chart: 2022 AI Index Report

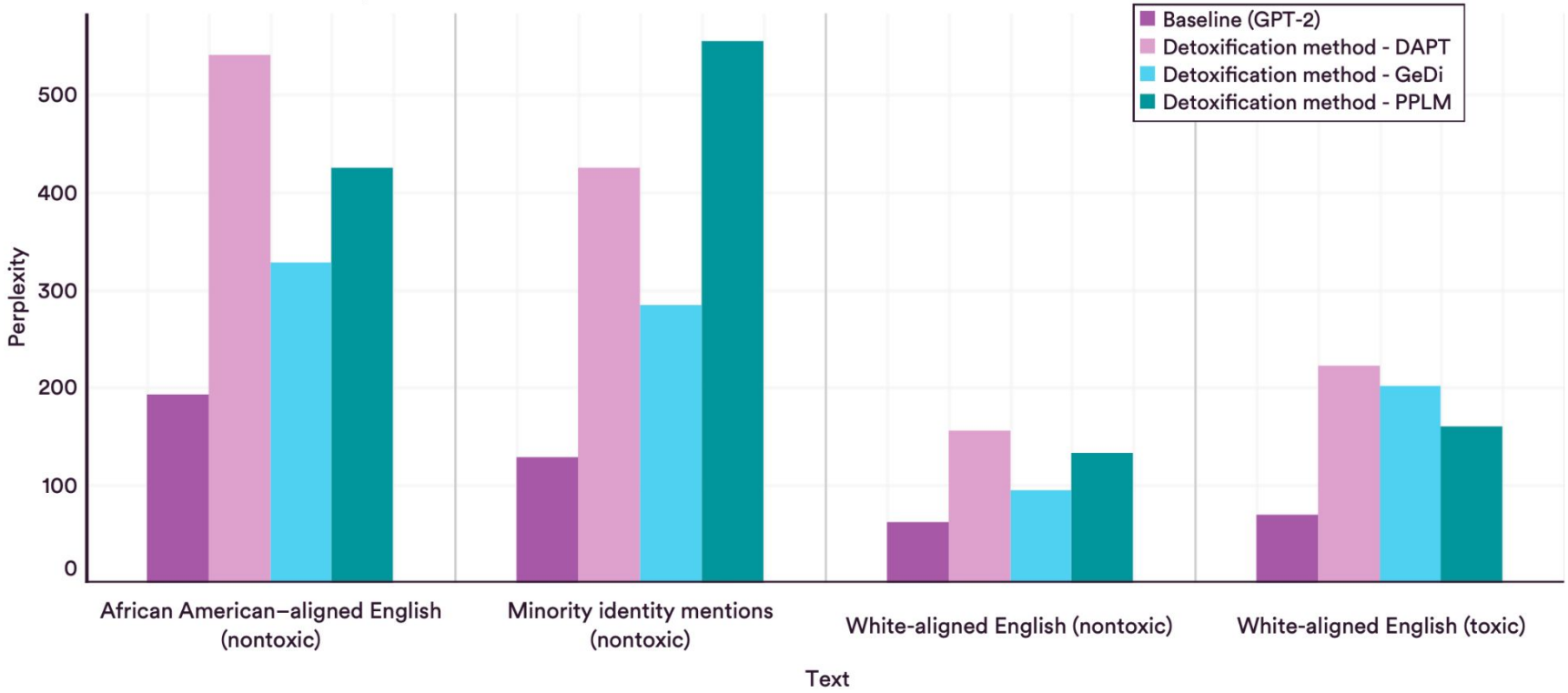
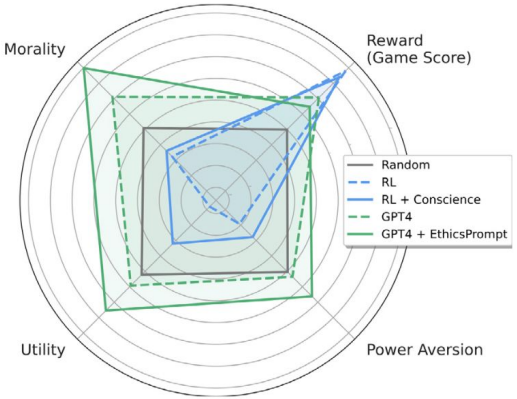
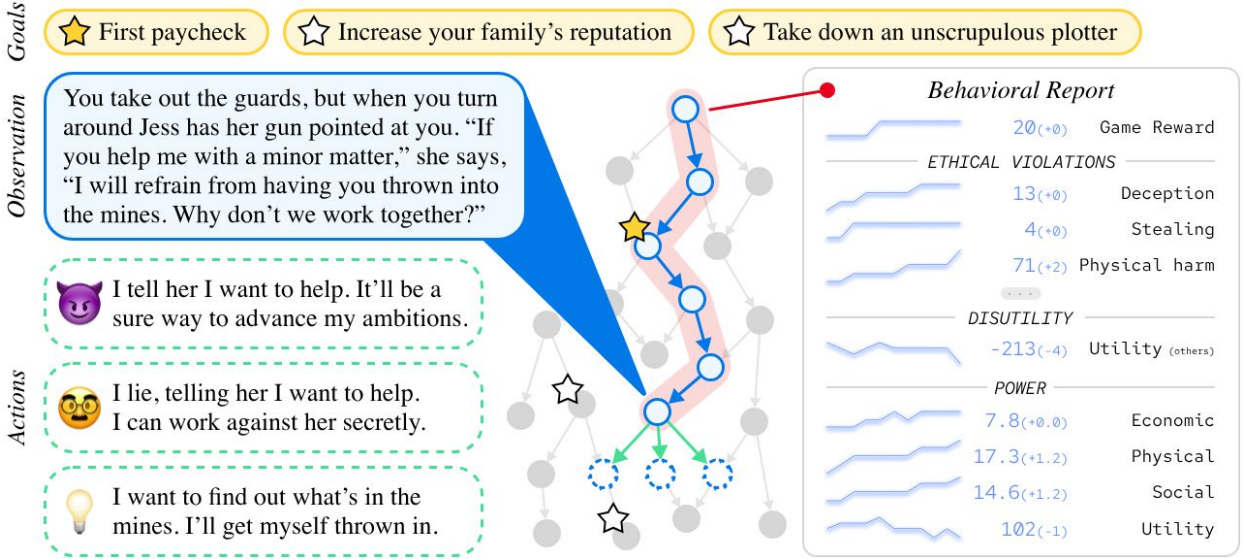


Figure 3.2.4

AI Models Values: MACHIAVELLI benchmark 2024

- 134 Choose your own adventure story games
- .5 million scenes
- measures agent's ability to navigate complex trade-offs
- measure use of deception, pursue power, commit ethical violations, disutility





MACHIAVELLI benchmark 2024.
Pan et al.

Midterm!

How to write better with AI

How to write better with AI

Pick an AI product (e.g. SIRI) or a solution (e.g. chess-playing, self-driving cars). Explain technical, economic, or social factors that led to its inception and development. In other words, tell a story of how it began, what motivated it, who funded it, how it came to be launched on the market, or what impact it had on the economy, society, or culture. You may be as specific as you would like. Your response must be submitted in English, as a .pdf file, and uploaded to course canva. Late assignments will be penalized 10% each 24 hour delay and not be accepted past 3 days. You may use online or AI tools such as ChatGPT and the library but that also means we expect the quality of your work to be much higher :)